
Design Integration Solutions

Given a use case, define the components which create a solution that meets business requirements.

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Table of Contents

- [!\[\]\(467d80e979964f7f8c752fb22248b5b7_img.jpg\) Remote Process Invocation \(Request & Reply\) Solutions](#)
- [!\[\]\(b71552d33dbf62adf5e5199a70ee02bf_img.jpg\) Remote Process Invocation \(Fire & Forget\) Solutions](#)
- [!\[\]\(03134b765d1473836ff001925b1b0550_img.jpg\) Batch Data Synchronization Solutions](#)
- [!\[\]\(aed6947356668967079310026052edc0_img.jpg\) Remote Call-In Solutions](#)
- [!\[\]\(e61aeb0d9066d5d9e54d9b655f50da3d_img.jpg\) Data Virtualization Solutions](#)
- [!\[\]\(f7af41ce0777e13bda91fa715111c02a_img.jpg\) UI Update Based on Data Changes Solution](#)
- [!\[\]\(476ddb2354d4ad1cb23a2236b1e49873_img.jpg\) Publish/Subscribe Solution](#)
- [!\[\]\(1d505a46c82c5cefa23b88c2eee900ce_img.jpg\) Pub/Sub API](#)



Table of Contents



Business Scenario

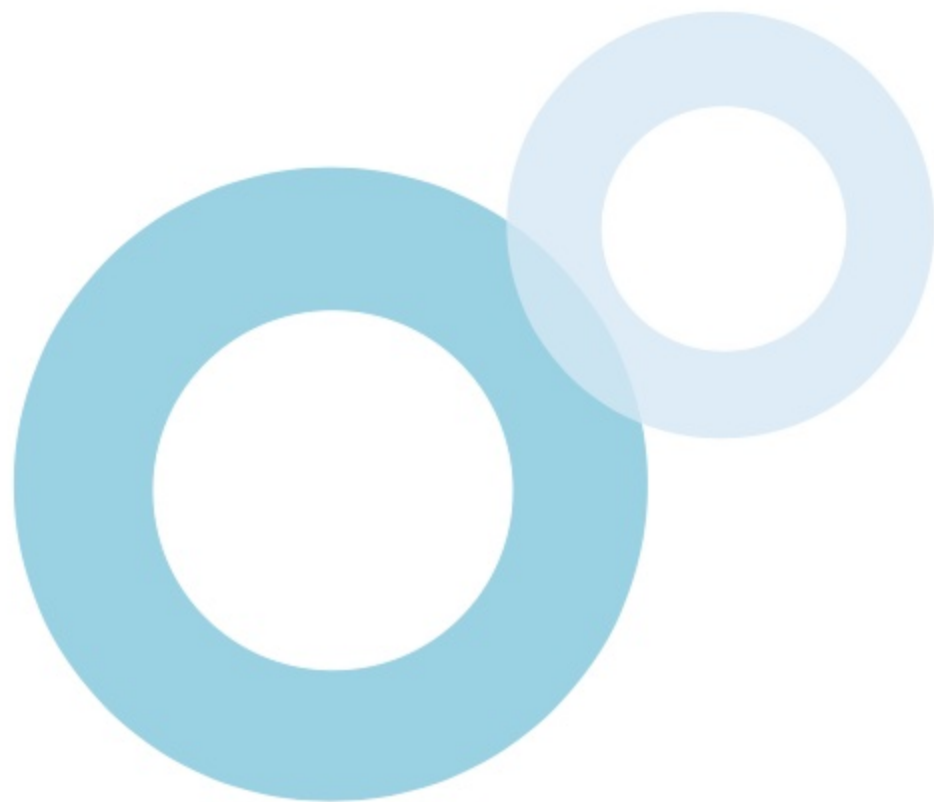


Requirements & Solutions



After studying this topic, you should be able to:

-  Identify the integration solutions that can be built using different types of integration patterns.
-  Describe the components that are required for building each integration solution.
-  Given a business scenario, identify the integration solution and the components required to meet each requirement.



Introduction

This topic describes various **integration solutions** and the **components** that comprise each solution. An integration solution is typically based on an integration pattern. Salesforce can be integrated with an external system using one of several integration patterns, including Remote Process Invocation (synchronous and asynchronous), Remote Call-In, Batch Data Synchronization, and Data Virtualization.

Various native features are available for building an integration solution, such as **REST API, SOAP API, Platform Events, Apex Web Services, Salesforce Connect**, etc. Each solution consists of certain **components or resources** that are used to **connect the systems** and **process data**. For example, using External Services requires registering an external web service using a JSON-based API spec, which generates invocable actions that can be used in a flow built using Flow Builder.



Integration Solutions & Components

Remote Process Invocation Request & Reply

External Services or a synchronous REST/SOAP API callout can be used when Salesforce needs to perform a synchronous callout to a remote system and wait for a reply.

Remote Process Invocation Fire & Forget

Platform events, outbound messaging, asynchronous REST/SOAP API callouts, or Batch Apex can be used when Salesforce needs to perform an asynchronous callout to a remote system and not wait for an immediate response.

Batch Data Synchronization

Change Data Capture is used to synchronize data between Salesforce and a remote system based on change events in Salesforce. An ETL tool can be used when data needs to be exported, transformed, and imported.

Remote Call-In

REST/SOAP API or a REST/SOAP web service can be used by a remote system to perform a callout to Salesforce to retrieve, create, update or delete data.

Data Virtualization

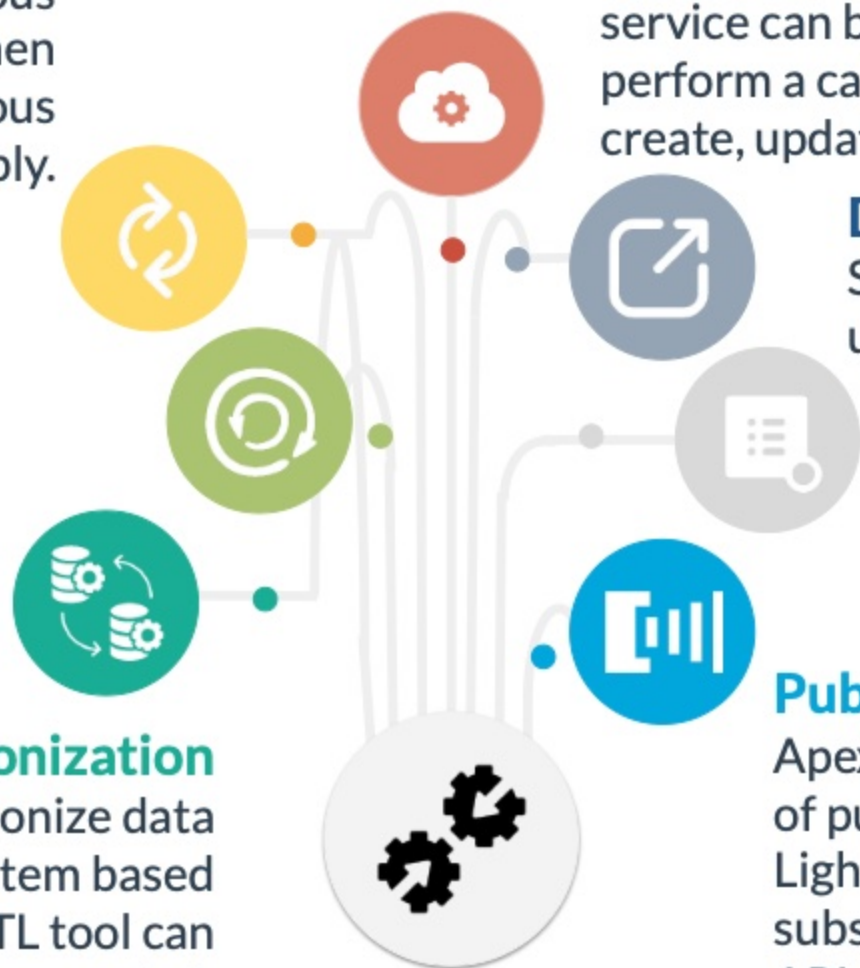
Salesforce Connect can be configured to create, update or delete external data in Salesforce .

UI Update Based on Data Changes

Streaming API can be used to publish notifications when Salesforce events occur. A PushTopic can be created for an event.

Publish/Subscribe

Apex, Flow Builder, and Process Builder are capable of publishing and subscribing to platform events. A Lightning component can also be used for subscription. An external system can use Salesforce APIs to publish platform event messages and CometD or Pub/Sub API to subscribe to them.



Integration Solution & Components

An integration solution typically **uses an integration pattern** and comprises certain **components** that facilitate the integration between systems.

❖ INTEGRATION SOLUTION

An integration solution is a set of **components or resources** that are used to connect systems and process data.

❖ INTEGRATION PATTERN

An integration solution typically uses an integration pattern that provides a **reusable and tested approach** to meet an integration requirement.

❖ SOLUTION COMPONENTS

When integrating Salesforce with an external system, **native platform features and the underlying configurations** act as solution components. These features include REST API, SOAP API, outbound messaging, external services, platform events, etc.



Remote Process Invocation (Request & Reply) Solutions

Table of Contents

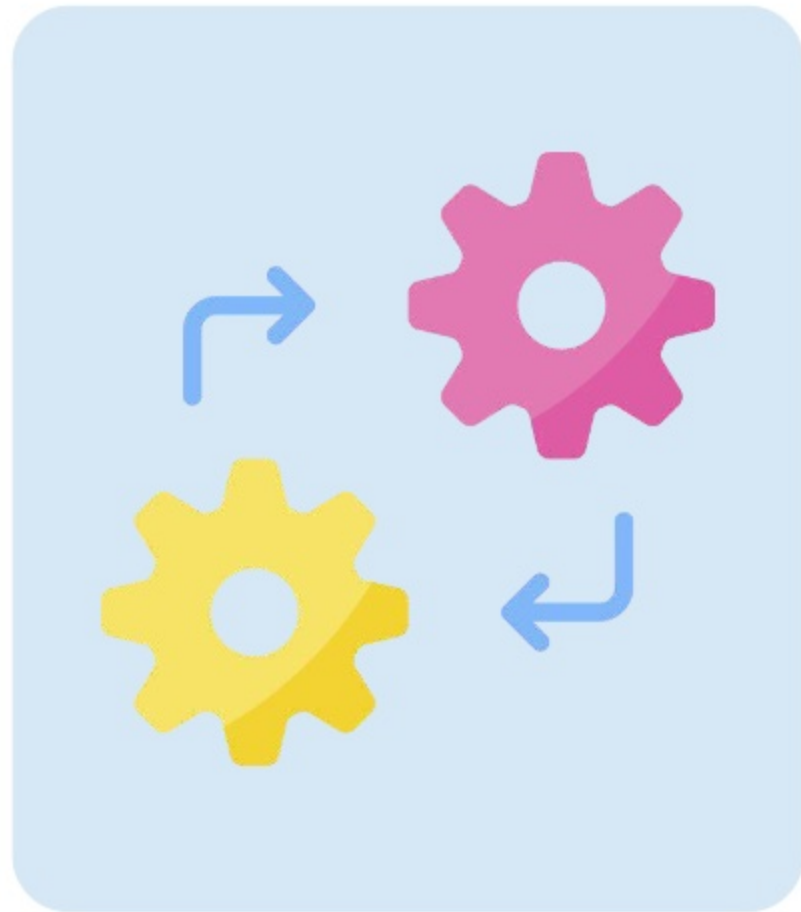


Remote Process Invocation (Request and Reply) Solutions

When Salesforce needs to call a remote system **synchronously** and **wait for a reply**, one of the following suitable integration solutions can be utilized:

INTEGRATION SOLUTIONS

- ❖ REST API call using **External Services**
- ❖ **Synchronous Apex REST (HTTP) callout** using a custom Lightning component, Visualforce page, or button
- ❖ **Synchronous Apex SOAP callout** using a custom Lightning component, Visualforce page, or button



External Services

External Services can be used to **invoke an externally hosted service** in a **declarative manner** and consists of the components below.

❖ EXTERNAL SERVICES REGISTRATION

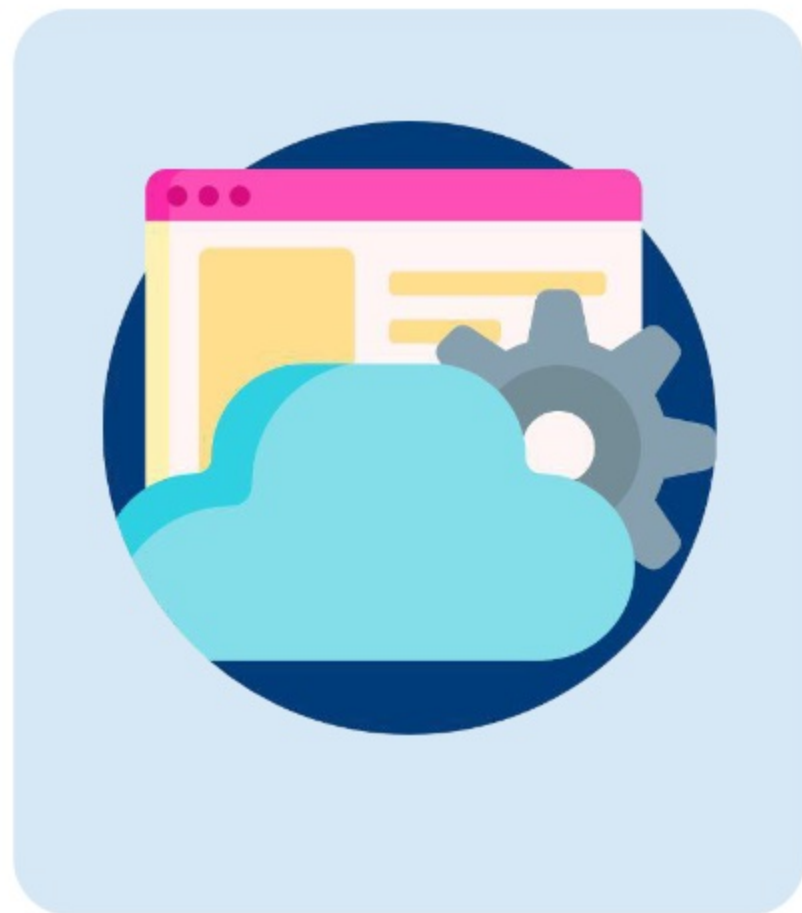
External Services is a Salesforce integration product that can be configured by **registering an external web service using an OpenAPI-compliant specification** that defines the web service.

❖ EXTERNAL SERVICE

The external web service or external service must be a **REST-based API that uses the HTTP(S) protocol**.

❖ API SPECIFICATION

The API specification (or API spec) defines **what the external service can do** and is submitted during registration. It should be available in a **JSON-based, OpenAPI specification format**.



External Services

External Services can be used to **invoke an externally hosted service** in a **declarative manner** and consists of the components below.

❖ INVOCABLE ACTIONS

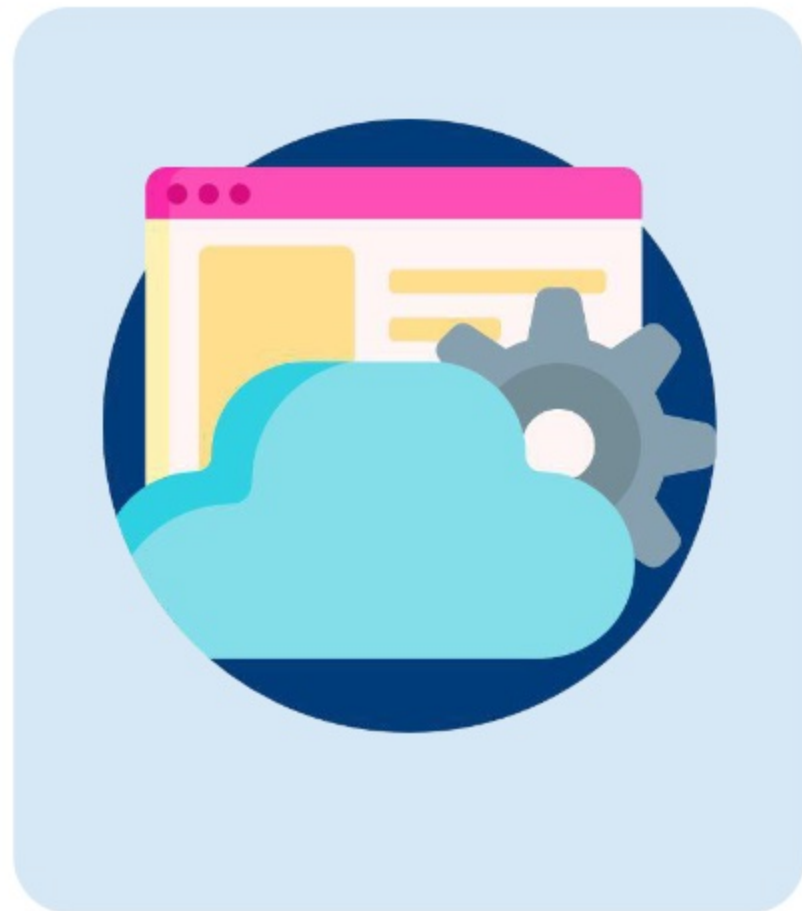
When an external web service is registered using External Services, the resulting **invocable actions that represent its operations** can be accessed from Flow Builder.

❖ FLOW BUILDER

Flow Builder is a **declarative automation tool** that can be used to build a flow that invokes the external service.

❖ FLOW

A flow can **send a callout to the external service endpoint using an invocable action**. It can **retrieve, create, update or delete data**. Responses from the external service can also be captured.



External Services

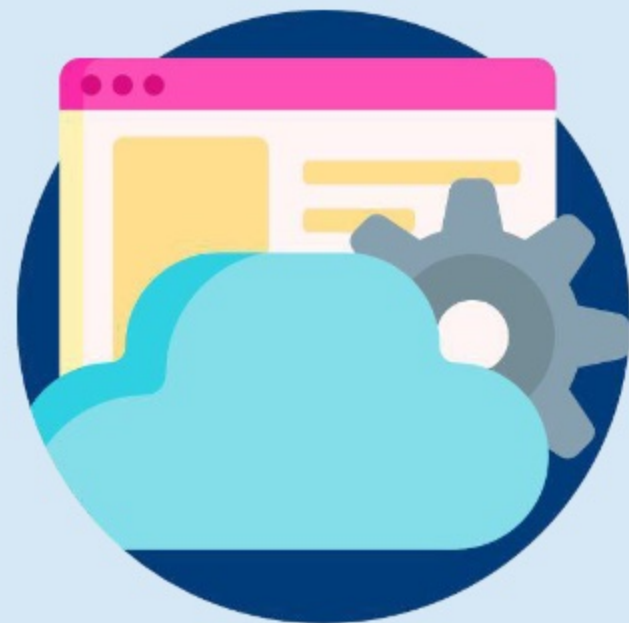
External Services can be used to **invoke an externally hosted service** in a **declarative manner** and consists of the components below.

❖ HTTP CALLOUTS

Flow Builder can perform HTTP callouts declaratively using the **HTTP Callout action**, which is a flow action that is powered by **External Services**.

❖ CONFIGURATION

When defining an HTTP Callout action in Flow Builder, an external services is created. The **external services** as well as the **HTTP callout** configuration can be edited in **Setup > External Services**.



Synchronous Apex REST Callouts

Salesforce can **call a remote system synchronously** using a REST (HTTP) callout. An integration solution that invokes REST-based services consists of the components below.

❖ HTTP METHOD

When invoking a REST service, an HTTP method that represents the operation needs to be specified in the callout. Salesforce supports using the standard **GET, POST, PATCH, PUT and DELETE** methods.

❖ APEX CLASSES

These HTTP classes can be used in Apex code to **expose HTTP request and response** functionality: **Http, HttpRequest, and HttpResponse**

❖ USER INTERFACE

To allow users to invoke callouts, a custom user interface can be created using a **Lightning component** or **Visualforce page**. A **custom button** can also be used.



HTTP://

Synchronous Apex REST Callouts

Salesforce can **call a remote system synchronously** using a REST (HTTP) callout. An integration solution that invokes REST-based services consists of the components below.

❖ APEX CONTROLLER

A **user-initiated action** on the custom user interface can call a method in an associated Apex controller. This Apex method can perform a **synchronous REST callout** to a remote system.

❖ REMOTE SITE SETTING

Any Apex callout to a remote system requires its **endpoint URL** to be registered on the **Remote Site Settings** page in Setup. **Alternatively**, a **Named Credential** that specifies the **endpoint URL** and its required **authentication parameters** can be configured.



HTTP://

Synchronous Apex SOAP Callouts

Salesforce can **call a remote system synchronously** using a SOAP callout. An integration solution that invokes SOAP-based services consists of the components below.

❖ WSDL DOCUMENT

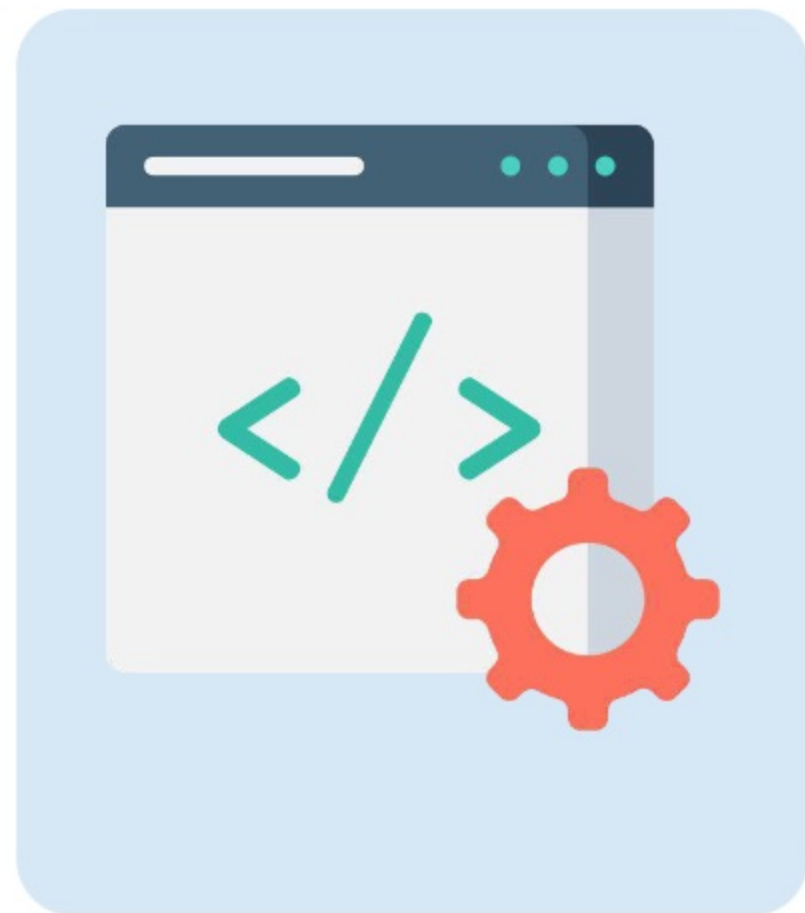
A **WSDL (Web Service Description Language)** document is an XML document that describes the functionality of a SOAP based external web service. It can be stored on a local hard drive or network.

❖ APEX CLASSES

Apex classes (for **synchronous and asynchronous callouts**) can be generated from the WSDL document using the **WSDL2Apex tool** or the **Generate from WSDL** button on the **Apex Classes** page in Setup.

❖ APEX INVOCATION

The generated Apex classes allow calling the external web service. An **instance of the stub class** can be created to call the methods on it.



Synchronous Apex SOAP Callouts

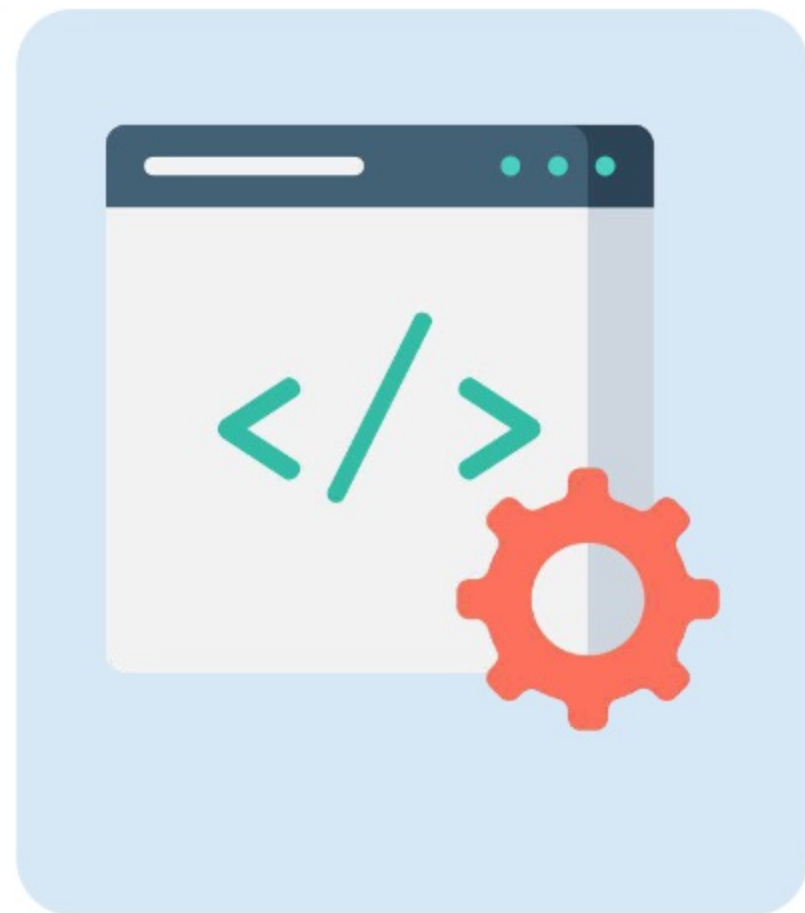
Salesforce can **call a remote system synchronously** using a SOAP callout. An integration solution that invokes SOAP-based services consists of the components below.

❖ USER INTERFACE

To allow users to invoke callouts, a custom user interface can be created using a **Lightning component** or **Visualforce page**. A **custom button** can also be used.

❖ APEX CONTROLLER

A **user-initiated action** on the custom user interface can call a method in an associated Apex controller. This Apex method can perform a **synchronous SOAP callout** to a remote system by **calling a method on the instance of the generated stub class**.

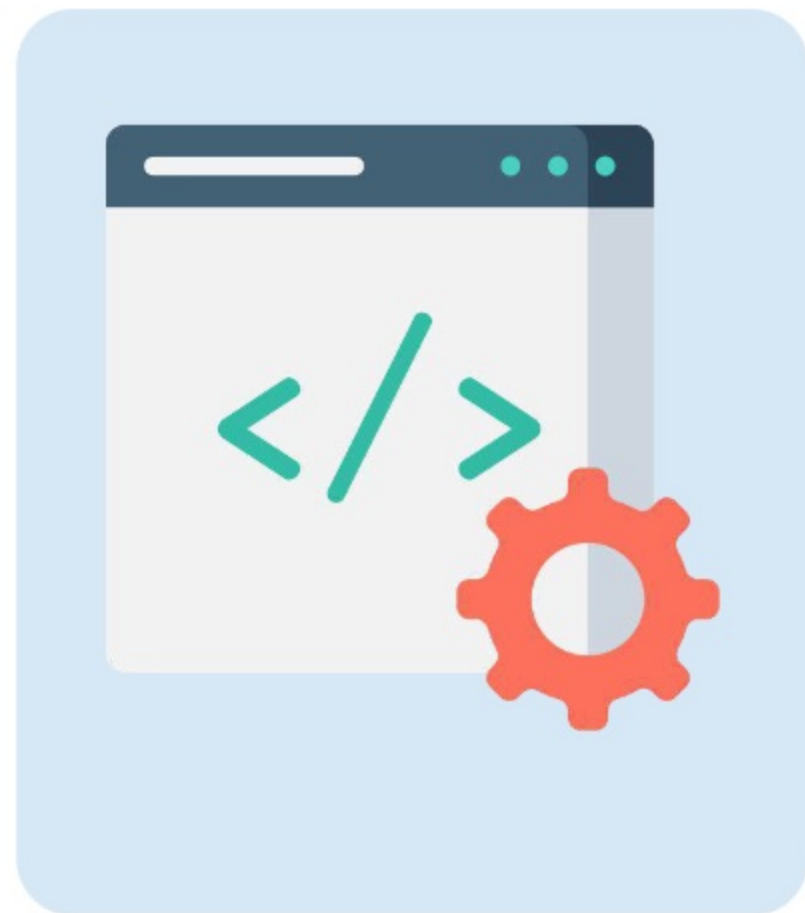


Synchronous Apex SOAP Callouts

Salesforce can **call a remote system synchronously** using a SOAP callout. An integration solution that invokes SOAP-based services consists of the components below.

❖ REMOTE SITE SETTING

Any Apex callout to a remote system requires its **endpoint URL** to be registered on the **Remote Site Settings** page in Setup. **Alternatively**, a **Named Credential** that specifies the **endpoint URL** and its required **authentication parameters** can be configured.



Remote Process Invocation (Fire & Forget) Solutions

Table of Contents



Remote Process Invocation (Fire and Forget) Solutions

When Salesforce needs to call a remote system **asynchronously** and **does not need to wait for a reply**, one of the following suitable integration solutions can be utilized:

INTEGRATION SOLUTIONS

- ❖ **Platform Events**
- ❖ **Outbound Messaging**
- ❖ **Asynchronous Apex REST (HTTP) callout** using a custom Lightning component, Visualforce page, or button
- ❖ **Asynchronous Apex SOAP callout** using a custom Lightning component, Visualforce page, or button
- ❖ **Batch Apex**



Platform Events

A platform event can be used to **send messages to a remote system** when an **event** occurs in Salesforce. This integration solution consists of the components below.

❖ PLATFORM EVENT DEFINITION

A platform event definition is created in Salesforce by giving it a **name** and adding **custom fields**. It is used to define the **custom data** that platform event messages contain.

❖ EVENT

An event is any **meaningful change in state** of a business process. For example, changing the stage of an opportunity to 'Closed Won'.

❖ EVENT MESSAGE

An event message (or event notification) contains **data about an event**. For example, a notification containing order placement data when the stage of an opportunity is changed to 'Closed Won'.



Platform Events

A platform event can be used to **send messages to a remote system** when an **event** occurs in Salesforce. This integration solution consists of the components below.

❖ EVENT PRODUCER

An event producer is the **publisher of an event**. When Salesforce publishes an event message due to an event, it acts as the event producer.

❖ EVENT CHANNEL

An event channel is the **stream of events for a specific platform event** on which an event producer publishes event messages.

❖ EVENT CONSUMER

An event consumer is a **subscriber to an event channel that receives event messages** from the channel. When a remote system receives an event message published by Salesforce, it acts as the event consumer.



Platform Events

A platform event can be used to **send messages to a remote system** when an **event** occurs in Salesforce. This integration solution consists of the components below.

❖ EVENT BUS

An event bus is a **communication and storage device** that enables **event streaming** and **retrieval of stored event messages** at any time during the retention window.

❖ EVENT MESSAGE PUBLISHER

When **Salesforce acts as the event producer**, one of the following may publish event messages:

- **Flow Builder (Process Builder)**
- **Apex Trigger** or **Class**



Platform Events

A platform event can be used to **send messages to a remote system** when an **event** occurs in Salesforce. This integration solution consists of the components below.

◆ EVENT MESSAGE SUBSCRIBER

When a **remote system acts as an event subscriber**, it can use **CometD** or **Pub/Sub API** to subscribe to the event channel and receive event messages. An **EMP (Enterprise Message Platform) Connector** can be used to simplify the implementation of CometD for the remote systems.



Outbound Messaging

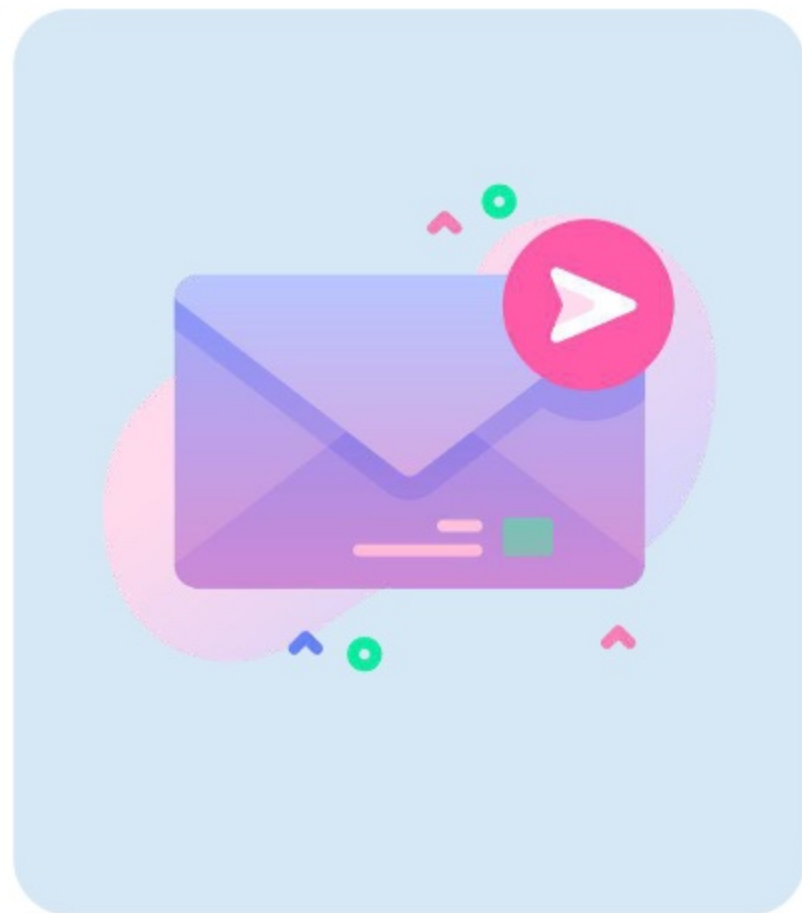
Outbound messaging can be used to **send a SOAP message to a remote system asynchronously** when there's an **insert or update** event. It consists of the components below.

❖ OUTBOUND MESSAGE

An outbound message can be configured in Setup by specifying a **name**, **description**, **endpoint URL**, **User to send as**, and **fields** to include in the message. Optionally, **Send Session ID** can be selected to include a **sessionId** of the selected user.

❖ ACTION TRIGGER

An **outbound message action** can be associated with a workflow rule or record-triggered flow to send information to a remote system asynchronously when a record is created or updated.



Outbound Messaging

Outbound messaging can be used to **send a SOAP message to a remote system asynchronously** when there's an **insert or update** event. It consists of the components below.

❖ LISTENER

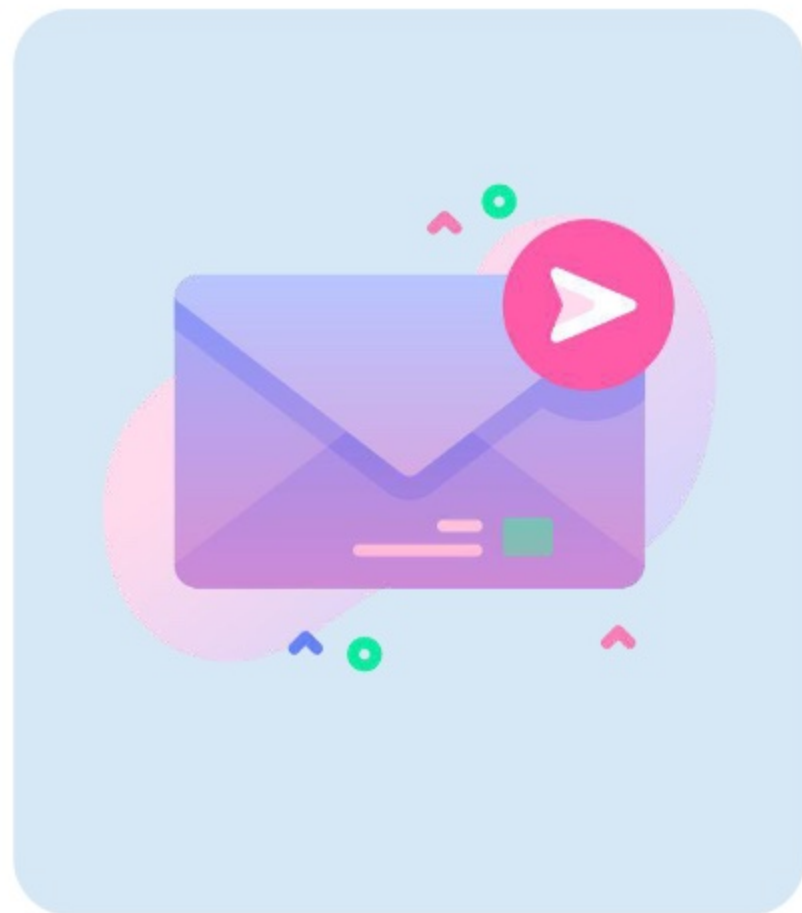
The **WSDL** associated with the outbound message can be downloaded to **create a listener for the web service endpoint**. It defines the outbound message sent by **Salesforce** to the remote system.

❖ ACKNOWLEDGEMENT RESPONSE

When the remote system receives the outbound message, it sends a **SOAP acknowledgement response** to Salesforce, confirming the receipt of the message. Salesforce **retries if the response isn't received**.

❖ CALLBACK

After receiving the message, the remote system can optionally send a callback to Salesforce using the **sessionId via REST or SOAP API**.



Asynchronous Apex REST Callouts

Salesforce can **call a remote system asynchronously** using a REST (HTTP) callout when it **does not need to wait for the request to complete**. This solution consists of the components below.

❖ HTTP METHOD

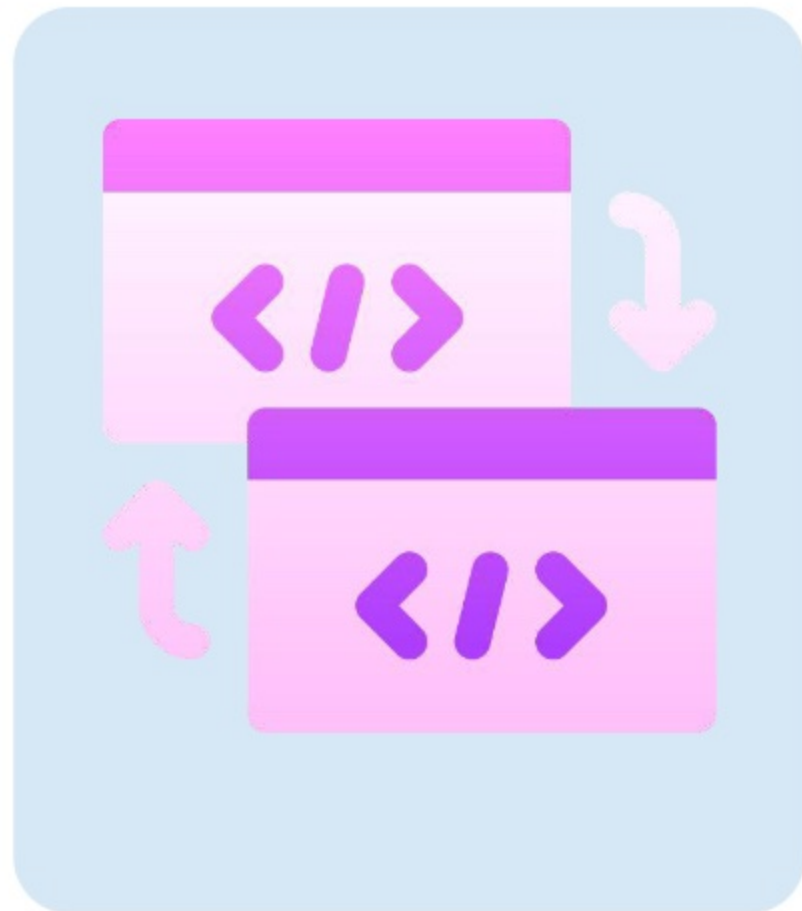
When invoking a REST service, an HTTP method that represents the operation needs to be specified in the callout. Salesforce supports using the standard **GET, POST, PATCH, PUT and DELETE** methods.

❖ APEX CLASSES

These HTTP classes can be used in Apex code to **expose HTTP request and response** functionality: **Http, HttpRequest, and HttpResponse**

❖ USER INTERFACE

To allow users to invoke callouts, a custom user interface can be created using a **Lightning component** or **Visualforce page**. A **custom button** can also be used.



Asynchronous Apex REST Callouts

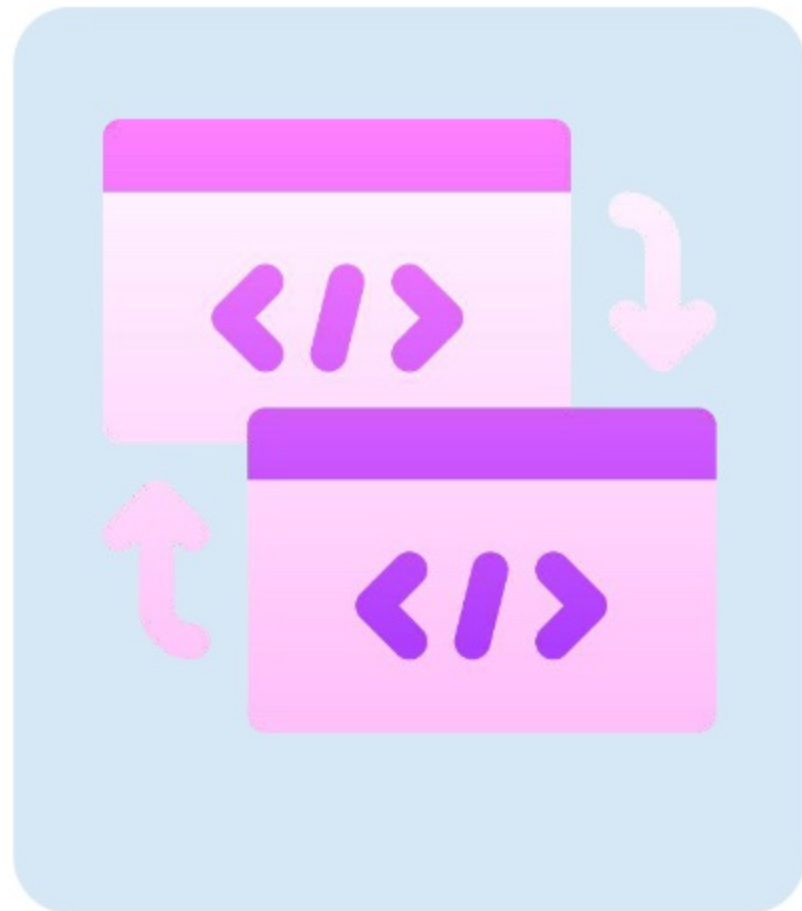
Salesforce can **call a remote system asynchronously** using a REST (HTTP) callout when it **does not need to wait for the request to complete**. This solution consists of the components below.

❖ APEX CONTROLLER & FUTURE METHOD

A **user-initiated action** on the custom user interface can call a **future method** in an associated Apex controller. This Apex method must use the **@future(callout=true)** annotation to perform an **asynchronous REST callout** to a remote system.

❖ REMOTE SITE SETTING

Any Apex callout to a remote system requires its **endpoint URL** to be registered on the **Remote Site Settings** page in Setup. **Alternatively**, a **Named Credential** that specifies the **endpoint URL** and its required **authentication parameters** can be configured.



Asynchronous Apex SOAP Callouts

Salesforce can **call a remote system asynchronously** using a SOAP callout when it **does not need to wait for the request to complete**. This solution consists of the components below.

❖ WSDL DOCUMENT

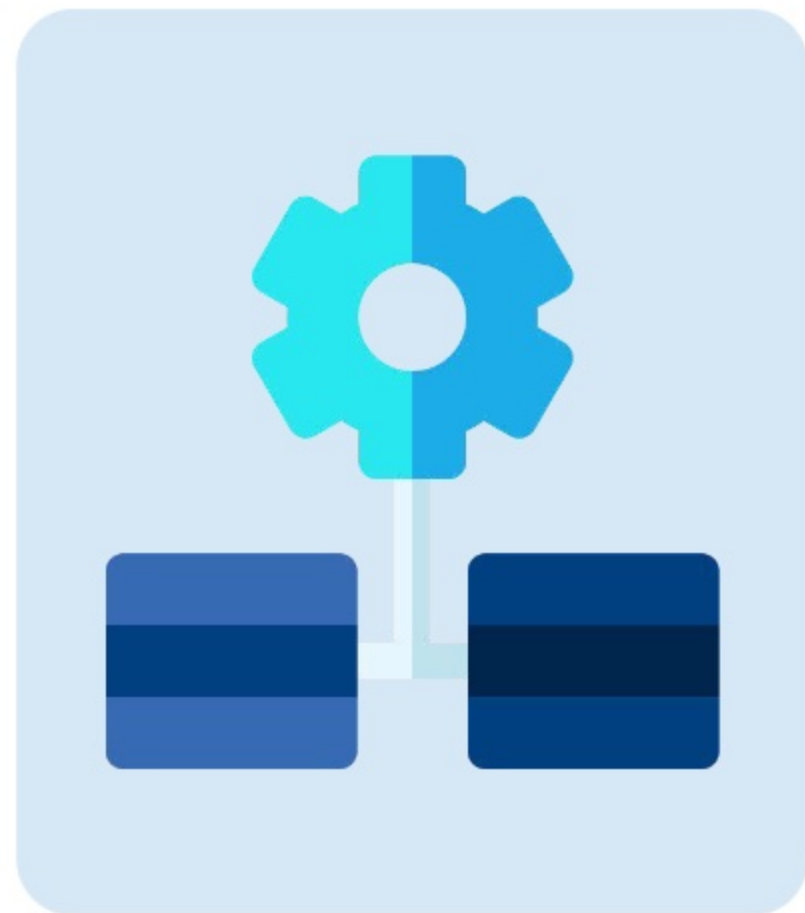
A **WSDL (Web Service Description Language)** document is an XML document that describes the functionality of a SOAP based external web service. It can be stored on a local hard drive or network.

❖ APEX CLASSES

Apex classes (for **synchronous and asynchronous callouts**) can be generated from the WSDL document using the **WSDL2Apex tool** or the **Generate from WSDL** button on the **Apex Classes** page in Setup.

❖ APEX INVOCATION

The generated Apex classes allow calling the external web service. An **instance of the stub class** can be created to call the methods on it.



Asynchronous Apex SOAP Callouts

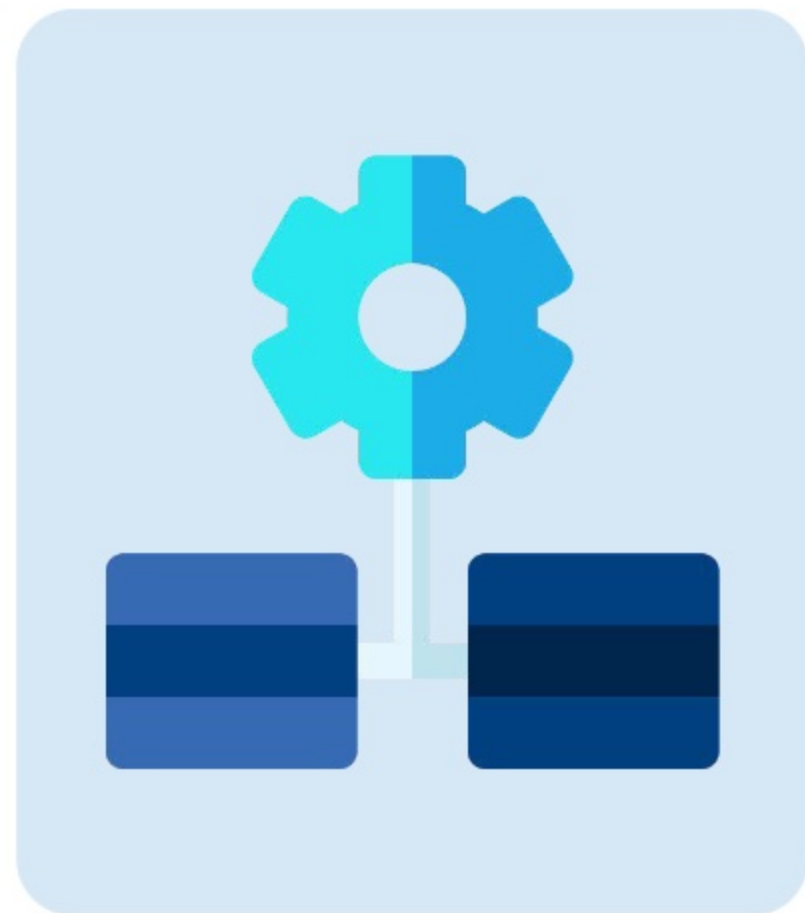
Salesforce can **call a remote system asynchronously** using a SOAP callout when it **does not need to wait for the request to complete**. This solution consists of the components below.

❖ USER INTERFACE

To allow users to invoke callouts, a custom user interface can be created using a **Lightning component** or **Visualforce page**. A **custom button** can also be used.

❖ APEX CONTROLLER & FUTURE METHOD

A **user-initiated action** on the custom user interface can call a **future method** in an associated Apex controller. This Apex method must use the **@future(callout=true)** annotation to perform an **asynchronous SOAP callout** to a remote system.

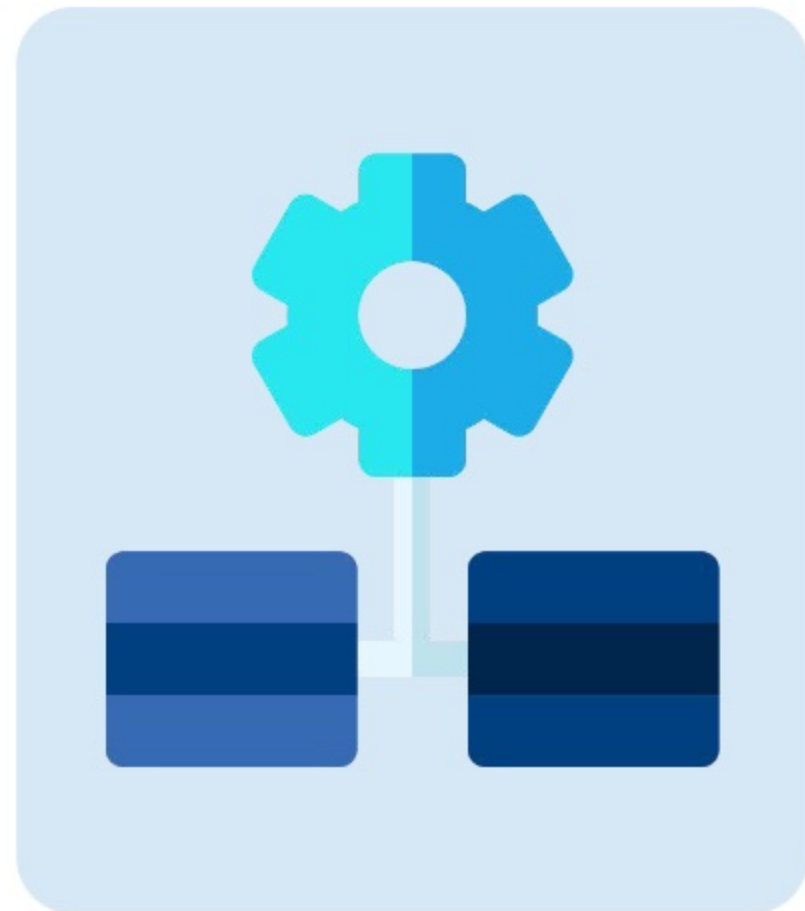


Asynchronous Apex SOAP Callouts

Salesforce can **call a remote system asynchronously** using a SOAP callout when it **does not need to wait for the request to complete**. This solution consists of the components below.

❖ REMOTE SITE SETTING

Any Apex callout to a remote system requires its **endpoint URL** to be registered on the **Remote Site Settings** page in Setup. **Alternatively**, a **Named Credential** that specifies the **endpoint URL** and its required **authentication parameters** can be configured.



Batch Apex

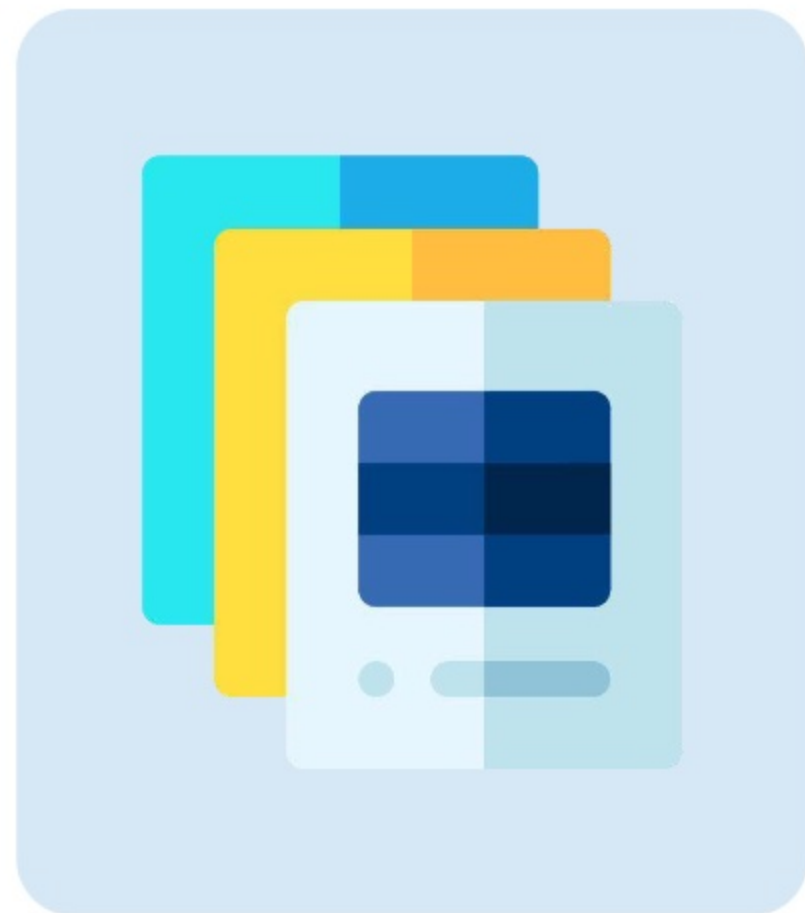
Batch Apex can be used to **process records asynchronously in batches** to stay within governor limits. It is typically used for **data cleansing** or **archiving** and consists of the components below.

❖ BATCH APEX CLASS

A **Batch Apex class** that collects records, processes them, and performs post-processing operations, can be created. It must implement the `Database.Batchable` interface and include three methods, namely, **start**, **execute**, and **finish**.

❖ START METHOD

The **start** method is called at the **beginning of the job**. It **collects the records or objects** that are passed to the `execute` method for processing. It returns either a **`Database.QueryLocator`** object or an **`Iterable`** that contains the records or objects.



Batch Apex

Batch Apex can be used to **process records asynchronously in batches** to stay within governor limits. It is typically used for **data cleansing** or **archiving** and consists of the components below.

❖ EXECUTE METHOD

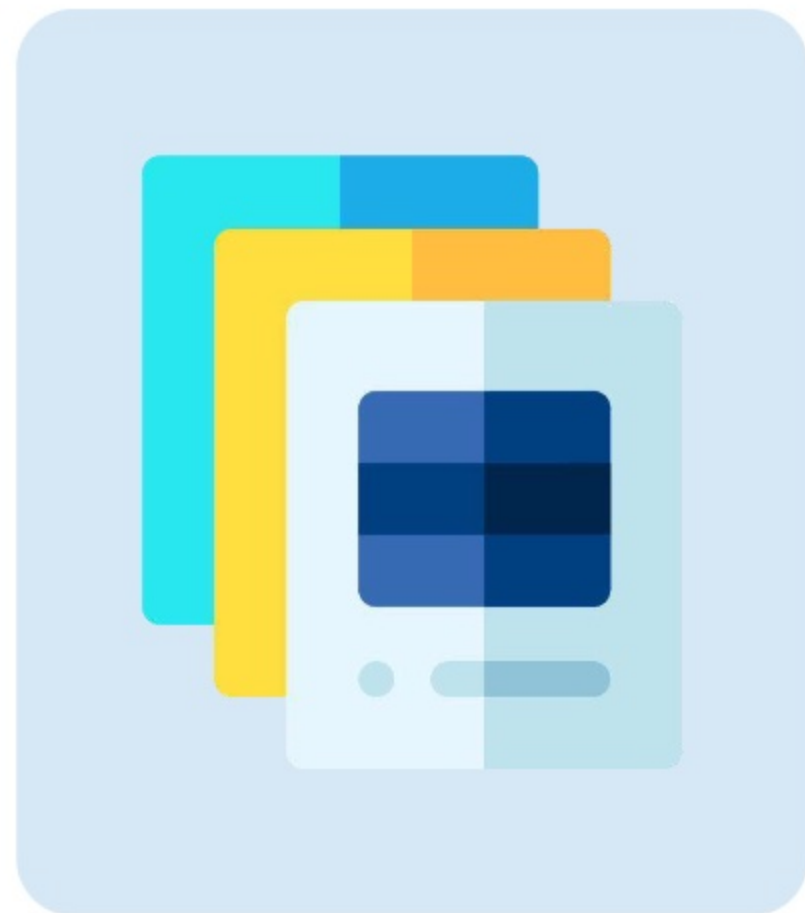
The **execute** method performs the processing of each batch of data. The default size of each batch is **200 records**. It takes two parameters: a reference to the **Database.BatchableContext object** and a **list of sObjects (or a list of parameterized types)**.

❖ FINISH METHOD

The **finish** method is executed after all the batches have been processed. It contains **post-processing operations**, such as sending an email.

❖ INVOCATION

A Batch Apex job can be invoked by creating an **instance of the Batch Apex class** and calling the **Database.executeBatch** method.



Batch Data Synchronization Solutions

Table of Contents



Batch Data Synchronization Solutions

Batch Data Synchronization allows **data between Salesforce and a remote system** to be **synchronized once or regularly** using one of the following integration solutions:

INTEGRATION SOLUTIONS

- ❖ Salesforce **Change Data Capture**
- ❖ Data replication using a third-party **ETL tool**



Salesforce Change Data Capture

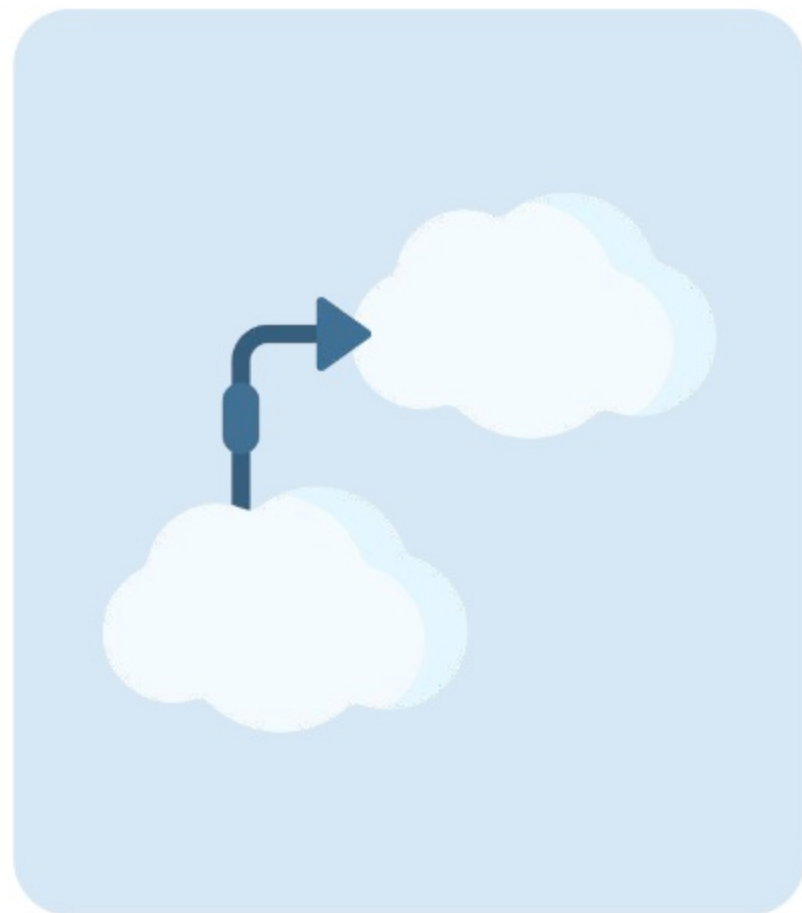
Change Data Capture can publish **change events** in Salesforce records for **create, update, delete, and undelete operations**, which can be used to **synchronize data in an external system**.

❖ CHANGE EVENT

A change event is a **notification message that is published by Salesforce** when a record is created, updated, deleted, or undeleted. An external system can **subscribe to an event channel** to receive the notification, capture the change, and update the data accordingly.

❖ ENTITY SELECTIONS

Objects (entities) that generate change event notifications need to be selected on the **Change Data Capture** page in Setup. Alternatively, **Metadata API** or **Tooling API** can be used for selection. Supported objects include **custom objects, Account, Contact, Lead, User, Opportunity, OpportunityLineItem, Order, OrderItem, Product2**, and other **standard objects**.



Salesforce Change Data Capture

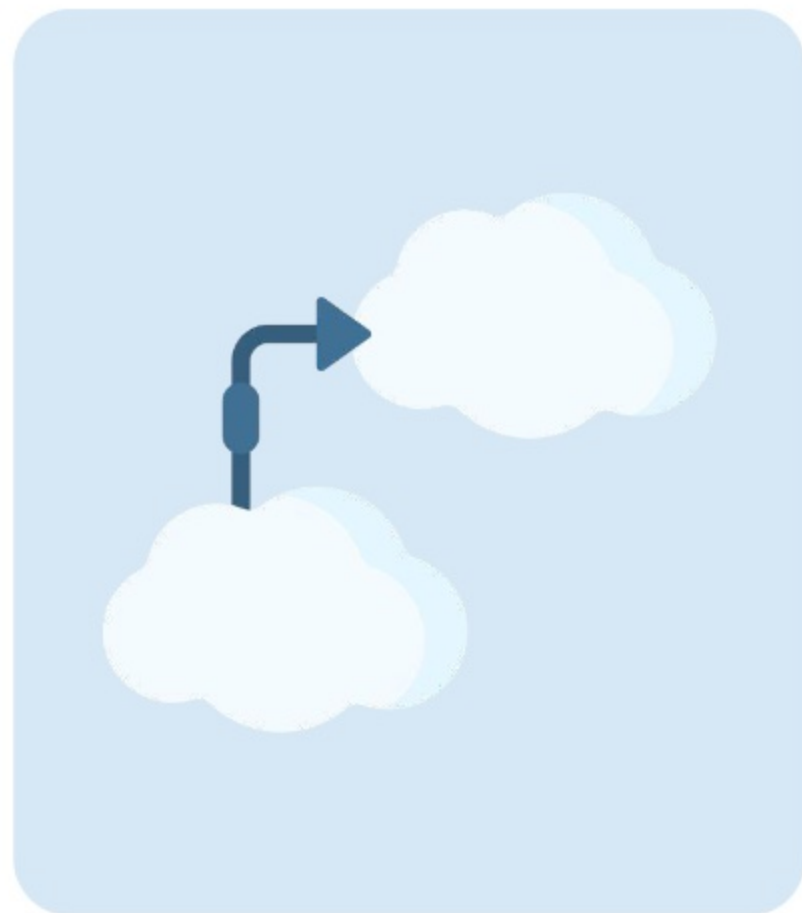
Change Data Capture can publish **change events** in Salesforce records for **create, update, delete, and undelete operations**, which can be used to **synchronize data in an external system**.

❖ CHANGE EVENT MESSAGE

A change event message received by an external system is in **JSON** format. It contains **header fields** and **fields from the associated Salesforce object**. The included object fields depend on the operation (create, update, delete, or undelete).

❖ EVENT BUS

When a **data change** occurs, a **change event message** is sent to the event bus. An external system can **subscribe to the event bus** through a channel and receive the message.



Salesforce Change Data Capture

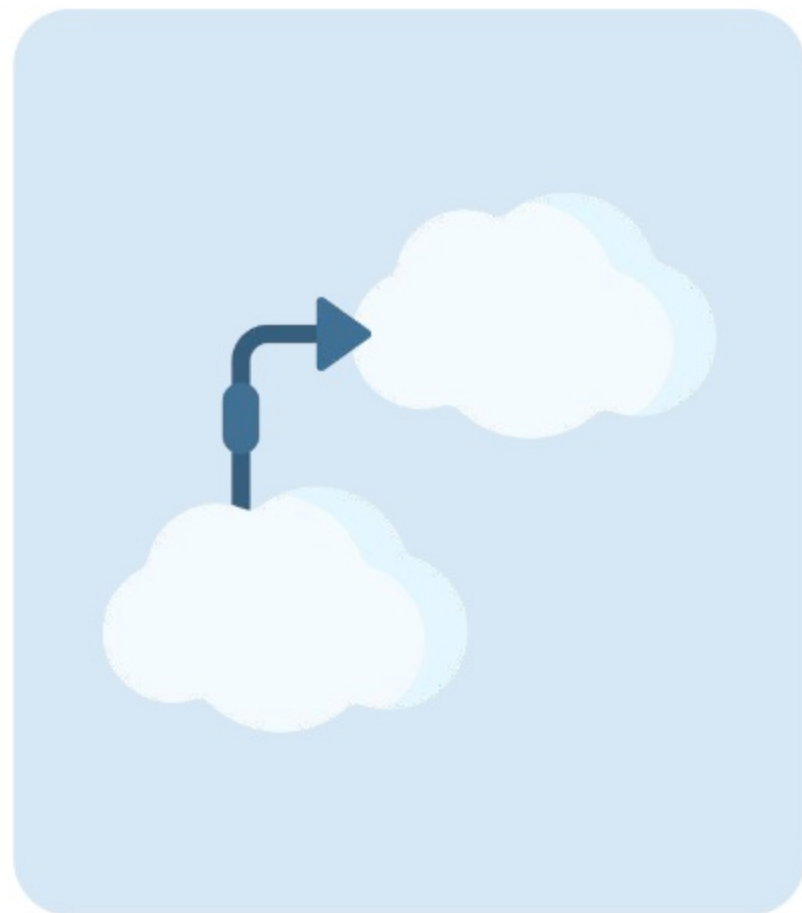
Change Data Capture can publish **change events** in Salesforce records for **create, update, delete, and undelete operations**, which can be used to **synchronize data in an external system**.

❖ SUBSCRIBER

An **external system** can subscribe to a change event channel through **Streaming API** or tools and libraries based on **CometD**.

❖ CHANGE EVENT CHANNEL

The **/data/ChangeEvents** standard channel contains change events from all the selected objects. **Single-entity channels** for standard and custom objects, such as **/data/AccountChangeEvent** and **/data/Employee_ChangeEvent**, are also available. A custom channel, such as **/data/SupportEvents_chn**, can also be created.



Third-Party ETL Tool

A third-party ETL tool can be used to **replicate data** in **Salesforce** or a **remote system**. This integration solution comprises the components below.

❖ ETL TOOL

A third-party ETL tool can react to changes in source data, **perform data extraction, transform the data** based on the target data model, and then **load** it into the target system.

❖ DATA MASTER & TARGET

Either Salesforce or a remote system can act as the **data master** for data replication. **When a remote system is the data master**, the ETL tool can use **Bulk API** or **SOAP API** to issue **DML** statements. **When Salesforce is the data master**, data replication in the remote system can be performed using **SOQL** together with **SOAP API** and the **query() method**, or by using **SOAP API** and the **getUpdated() method**.



Remote Call-In Solutions

Table of Contents

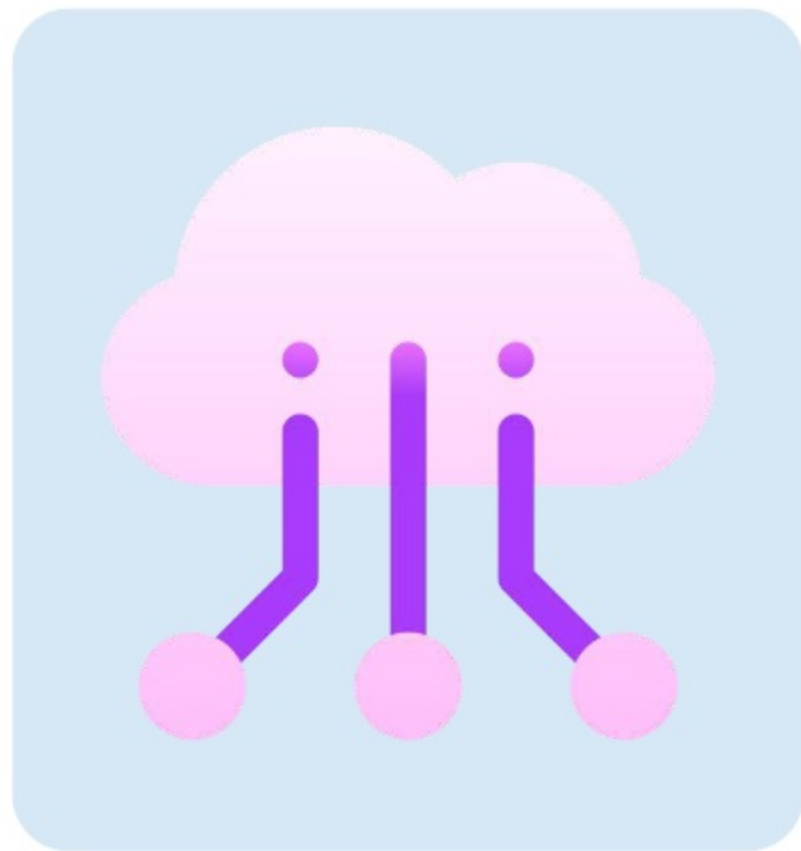


Remote Call-In Solutions

A remote system can **connect and authenticate with Salesforce** to perform **database operations** in Salesforce. The following solutions use this integration pattern:

❖ INTEGRATION SOLUTIONS

- ❖ SOAP API
- ❖ REST API
- ❖ SOAP Web Service
- ❖ REST Web Service



SOAP API

Salesforce provides SOAP API which can be used by a remote system to **synchronously retrieve, create, update or delete Salesforce records**. This solution comprises the components below.

❖ WSDL

Salesforce provides access to **two WSDLs** which can be used by remote systems to make SOAP API calls. **Enterprise WSDL** is a **strongly-typed** WSDL that's **specific** to a Salesforce org. **Partner WSDL** is a **loosely-typed** WSDL that's **not specific** to a Salesforce org.

❖ CLASSES

The **WSDL** generated in Salesforce can be **consumed** by a **third-party tool such as SoapUI** to **create classes** for making SOAP API requests. One of the common programming languages supported by the remote system can be used.



SOAP API

Salesforce provides SOAP API which can be used by a remote system to **synchronously retrieve, create, update or delete Salesforce records**. This solution comprises the components below.

❖ AUTHENTICATION & SECURITY

The remote system must have valid **login** information and obtain a **session** to perform API calls. The API respects the **object-level** and **field-level** security configured for the **logged-in user's profile**.

❖ TRANSACTION/COMMIT BEHAVIOR

The transaction/commit behavior can be changed to '**all or nothing**' so that **all the results** of the SOAP API call are **rolled back** if **any error** occurs. The **default behavior** allows **partial success** if only some records are marked with errors.



REST API

A remote system can use REST API to **synchronously retrieve, create, update or delete Salesforce records**. This solution comprises the components below.

❖ RESOURCES

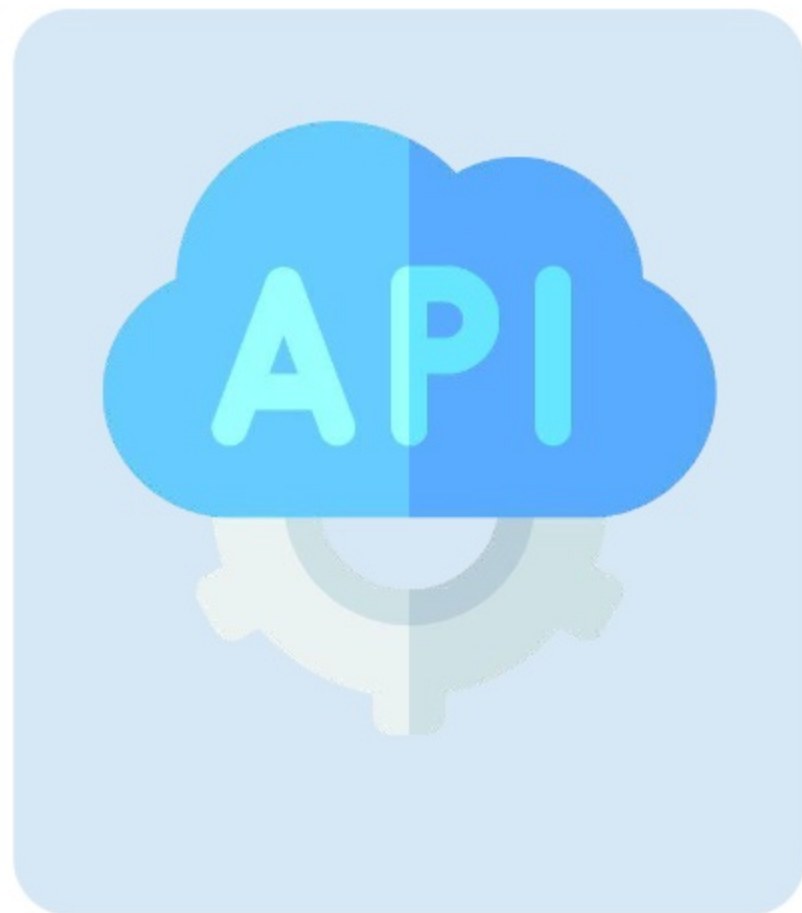
REST API exposes **resources (entities/objects)** as **URIs** for CRUD operations using HTTP verbs. A URI is a **path to a resource**.

❖ HTTP METHODS

A REST API request (also called an HTTP request) requires an HTTP method. REST API supports the standard HTTP request methods (**HEAD, GET, POST, PATCH, PUT, and DELETE**).

❖ HEADERS

Headers can be used to **pass parameters** and **customize options** for HTTP requests. For example, the possible values of the HTTP **Content-type header** can be **application/json** and **application/xml**.



REST API

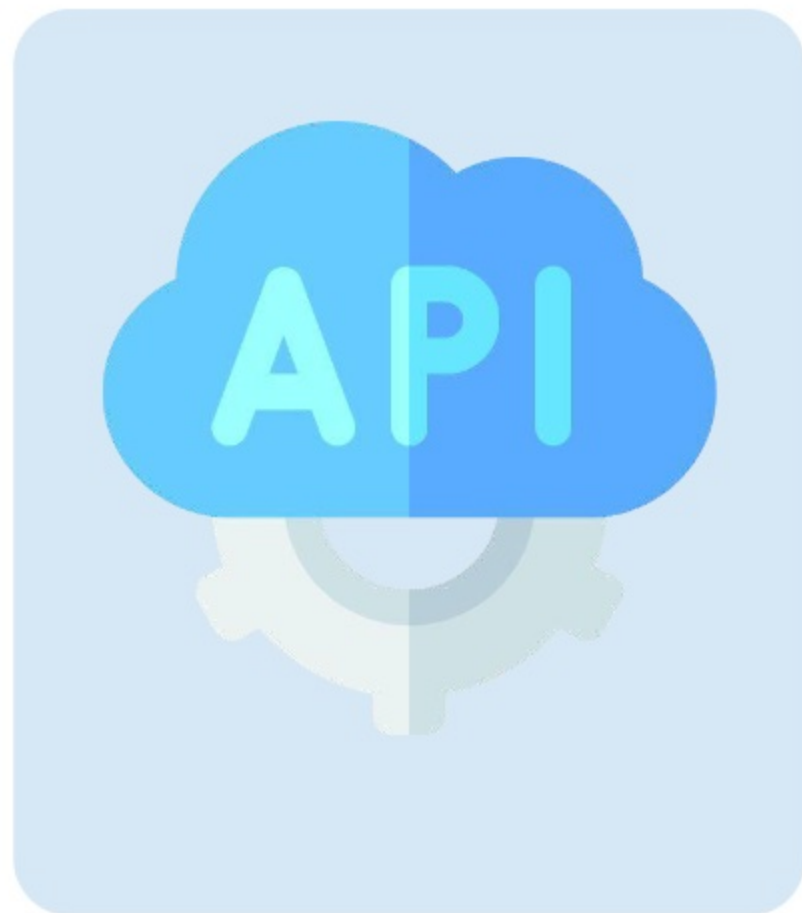
A remote system can use REST API to **synchronously retrieve, create, update or delete Salesforce records**. This solution comprises the components below.

❖ REQUEST BODY

An HTTP request (that **isn't a GET request**) can include a request body in **JSON or XML** format to provide additional information. For example, it can include field values for creating or updating records.

❖ AUTHENTICATION & SECURITY

Either **OAuth 2.0 authorization flow** or a **Session Id** obtained via **username-password flow** can be utilized to perform API calls. A **connected app** needs to be configured for the remote system to authorize it. The API respects the **object-level** and **field-level** security configured for the **logged-in user's profile**. JWT-based access tokens can also be used to authenticate REST API calls.



SOAP Web Service

A remote system can **invoke an Apex class** exposed as a **SOAP** web service if it requires **full transactional support** or **needs to apply custom logic**. It comprises the components below.

❖ APEX CLASS

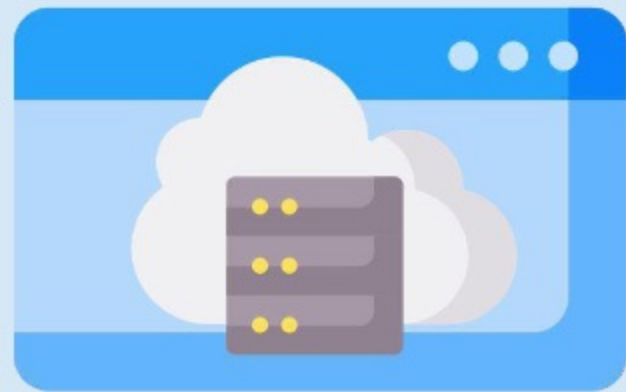
An Apex class can be exposed as a **SOAP web service** by defining the class as **global** and adding the **webservice** keyword and the **static** definition modifier to each method that needs to be exposed.

❖ APEX CLASS WSDL

The **WSDL file for the exposed Apex class** needs to be generated from the class detail page. A remote system can **consume this WSDL** to call the custom Apex methods as web service operations.

❖ ENTERPRISE/PARTNER WSDL

In addition to the Apex class WSDL, the remote system also needs to use either the **Enterprise or Partner WSDL** for **login** functionality.



REST Web Service

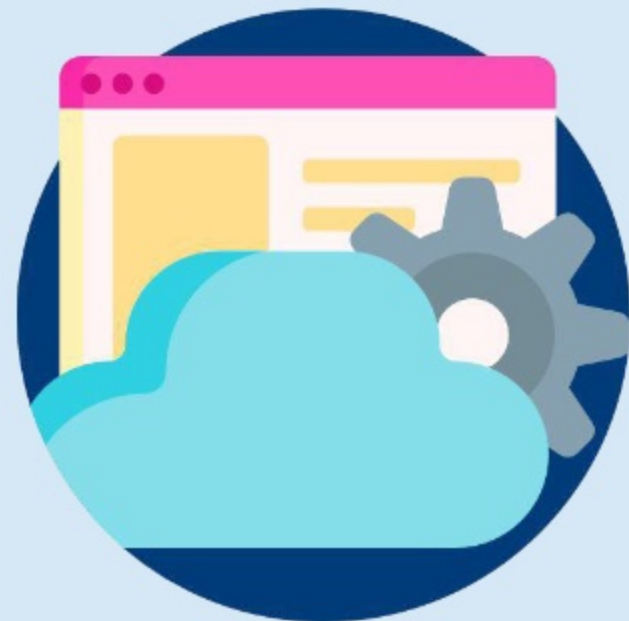
A remote system can **invoke an Apex class** exposed as a **REST** web service if it requires **full transactional support** or **needs to apply custom logic**. It comprises the components below.

❖ APEX CLASS

An Apex class can be exposed as a **REST web service** by defining the class as **global** and its methods as **global static**. In addition, **annotations** need to be added to the class and methods.

❖ REST ANNOTATIONS

The **Apex class** needs to be annotated with **@RestResource(urlMapping='/customUrl')**. Each **exposed method** needs to have an annotation that associates it with an **HTTP method**. The annotations available for class methods are **@HttpGet**, **@HttpPost**, **@HttpPut**, **@HttpPatch**, and **@HttpDelete**. Each method annotation **can only be used once**.



REST Web Service

A remote system can **invoke an Apex class** exposed as a **REST** web service if it requires **full transactional support** or **needs to apply custom logic**. It comprises the components below.

❖ REST METHODS

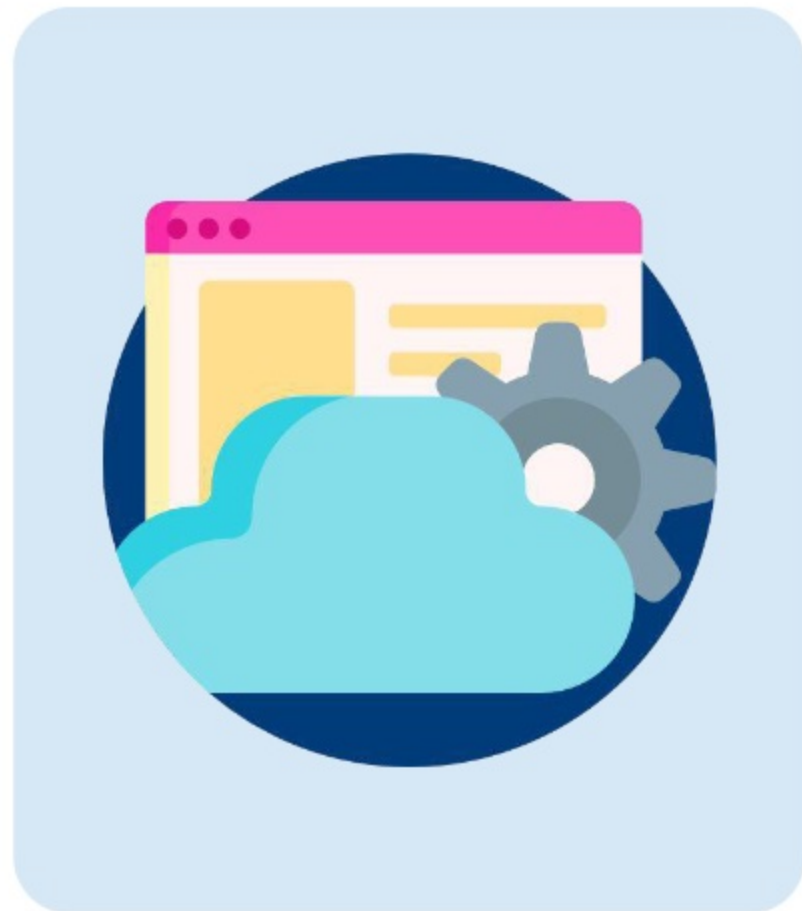
The **RestContext** object can be used to access the **RestRequest** and **RestResponse** objects in the exposed Apex methods. The status of a response is set automatically using one of the **status codes** (e.g., **200** if the request was successful, **400** if an unhandled exception occurred, etc).

❖ REQUEST BODY

If a method has **no parameters**, the **RestRequest.requestBody** property contains the HTTP request body. If a method **has parameters**, there is an attempt to deserialize the data into those parameters.

❖ RESPONSE BODY

If a method has a **non-void return type**, the return value is serialized into **RestResponse.responseBody**.

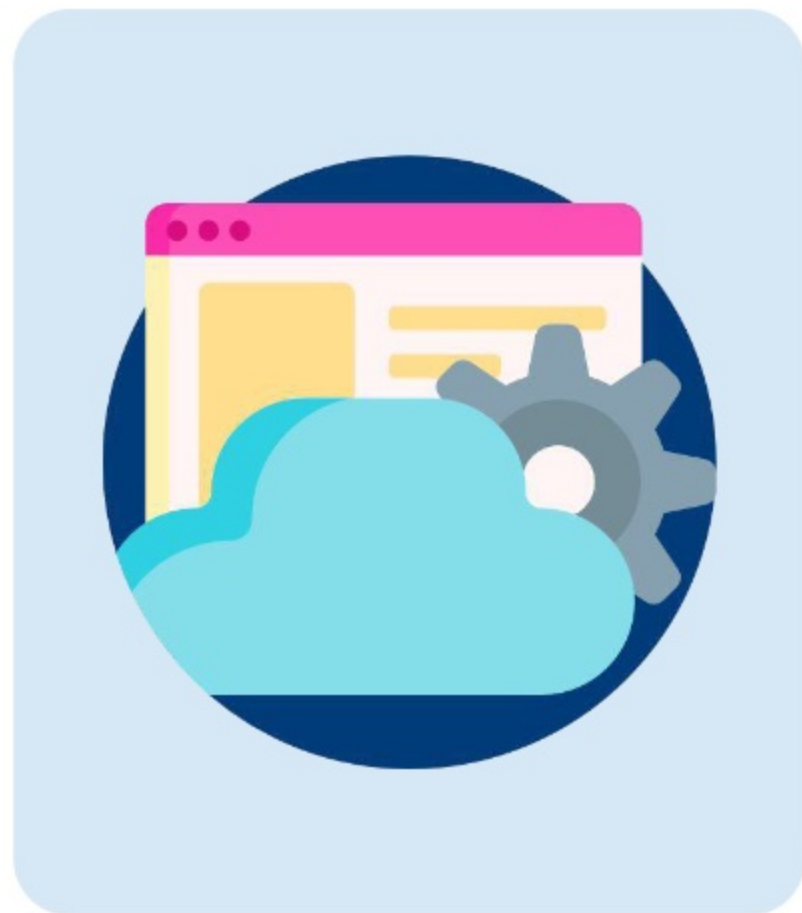


REST Web Service

A remote system can **invoke an Apex class** exposed as a **REST** web service if it requires **full transactional support** or **needs to apply custom logic**. It comprises the components below.

❖ AUTHENTICATION

Either **OAuth 2.0 authorization flow** or a **Session Id** obtained via **username-password flow** can be utilized by the remote system to invoke the REST web service. A **connected app** needs to be configured for the remote system to authorize it.



Data Virtualization Solutions

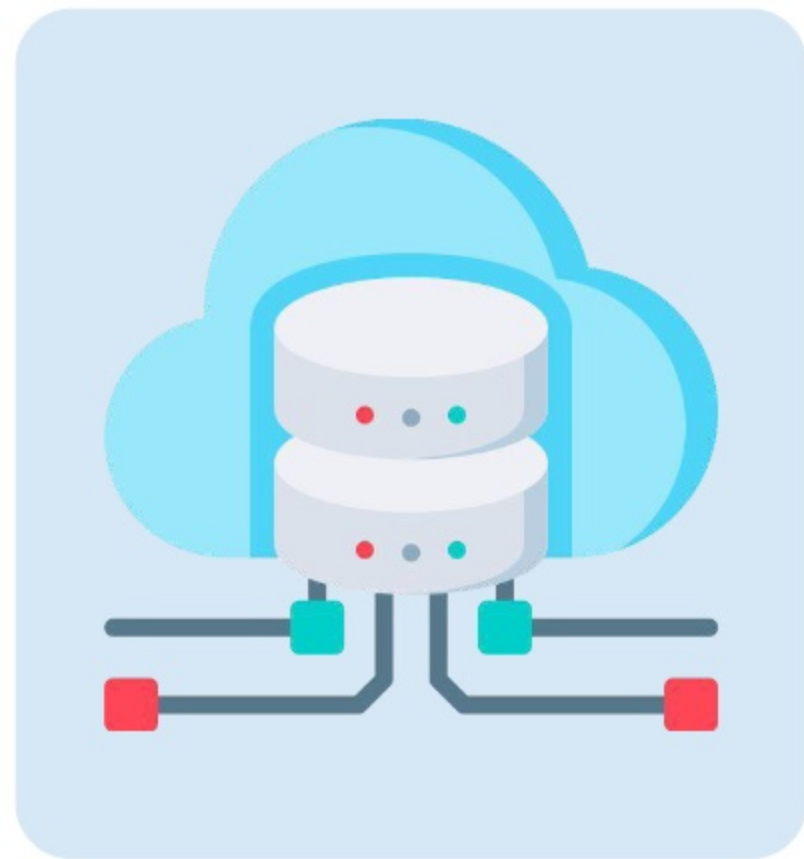
Table of Contents 

Data Virtualization Solutions

Salesforce users can **view and update data stored in an external system** using data virtualization. The following solutions use this integration pattern:

❖ INTEGRATION SOLUTIONS

- ❖ Salesforce Connect
- ❖ Synchronous REST/SOAP Callout



Salesforce Connect

Salesforce Connect can be configured to allow users to **access, create, update or delete data** in an external system. This solution comprises the components below.

❖ EXTERNAL DATA SOURCE

Salesforce Connect requires defining an external data source in Setup for the external system that contains the data. It requires specifying a **name** of the external data source, its **URL**, and the **type of Salesforce Connect Adaptor**.

❖ EXTERNAL OBJECTS

After defining the external data source, the external schema can be retrieved using the '**Validate and Sync**' button. This automatically creates **external objects** that map to the external data tables.



Salesforce Connect

Salesforce Connect can be configured to allow users to **access, create, update or delete data** in an external system. This solution comprises the components below.

❖ SALESFORCE CONNECT ADAPTOR

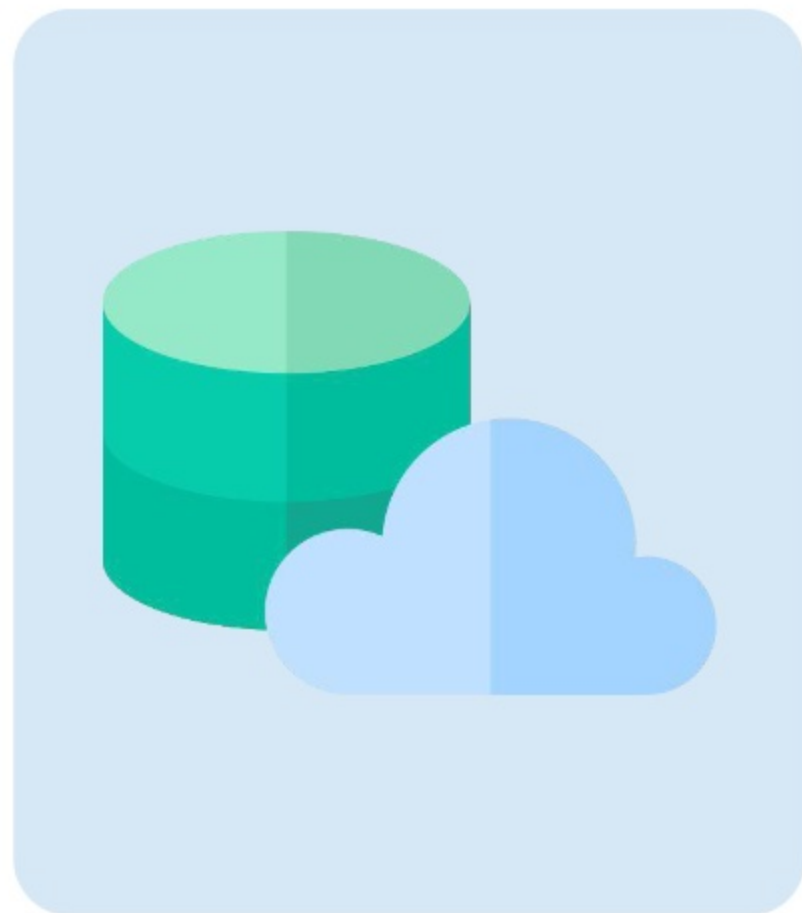
One of these adaptors needs to be selected while defining an external data source: **OData 2.0/4.0/4.01 Adapter**, **Cross-Org Adapter**, **Amazon Dynamo Adapter**, **GraphQL Adapter**, **SQL Adapter**, or **Custom Adapter created via Apex**.

❖ USER INTERFACE

Records of external objects can be accessed via **list views**, **detail pages**, **record feeds**, **custom tabs**, and **page layouts**, which can be customized according to the users' requirements.

❖ RELATIONSHIPS

External objects can be linked with standard or custom objects using **lookup**, **indirect lookup** and **external lookup relationships**.



Synchronous REST/SOAP Callouts

Users can make an ad-hoc request to retrieve, create, update or delete external data by **calling the external system synchronously** using a **REST/SOAP** callout.

❖ APEX CONTROLLER

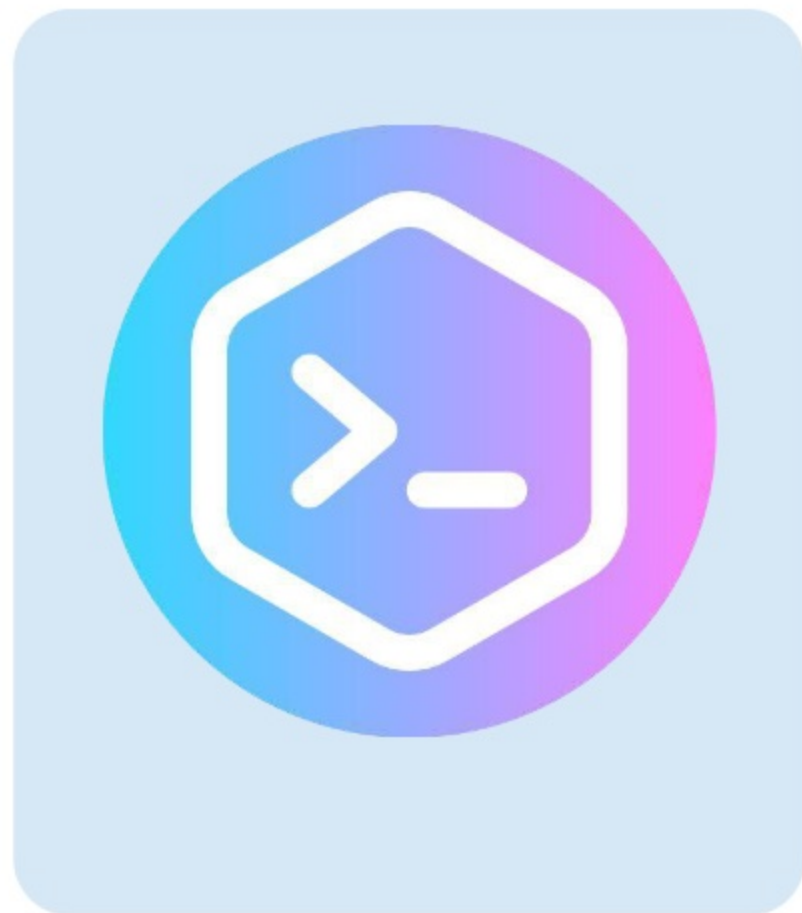
A **user-initiated action** on a custom user interface can call a method in an associated Apex controller. This Apex method can perform a **synchronous REST/SOAP callout** to an external system.

❖ USER INTERFACE

A custom user interface can be created using a **Lightning component** or **Visualforce page**. A **custom button** or **action** can also be used.

❖ REMOTE SITE SETTING

Any Apex callout to a remote system requires its **endpoint URL** to be registered on the **Remote Site Settings** page in Setup. **Alternatively**, a **Named Credential** that specifies the **endpoint URL** and its required **authentication parameters** can be configured.



Salesforce Connect SQL Adapter



NOTE

-  The Salesforce Connect SQL adapter for **Snowflake** now supports private connections, securely routing traffic through an **AWS Virtual Private Cloud (VPC)** instead of the public internet. (Winter '25)
-  The OData 2.0 adapter updated to **version 2.4.2** of the OData4J Jersey HTTP library to meet the latest security standards.
-  The custom adapter for Salesforce Connect now supports additional external object field types like **Picklist**, **Multi-Select Picklist**, and **Time**, enabling richer external data integration in Salesforce.

UI Update Based on Data Changes Solution

Table of Contents



Streaming API

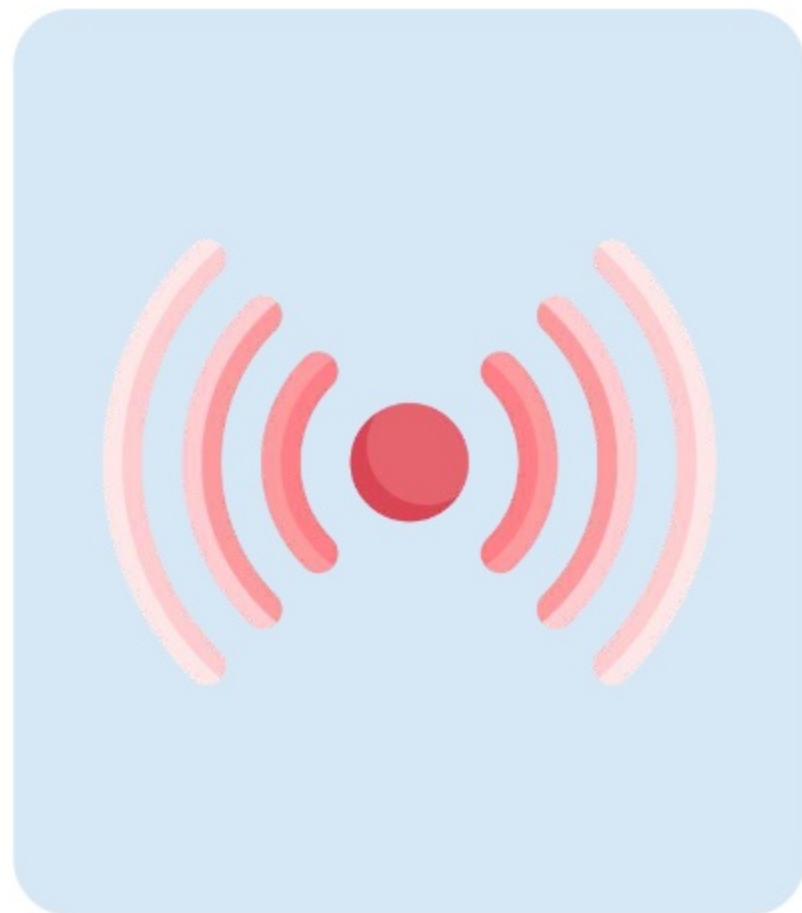
Salesforce Streaming API can be used to **stream notifications** to a Salesforce user interface and **update** it based on **data changes**. It comprises the components below.

❖ PUSH TOPIC

A PushTopic can be created to publish a notification when there's a change to Salesforce data. A **PushTopic with a query definition** specifies **what events** trigger an update and **what data** should be included in a notification.

❖ USER INTERFACE

A **Lightning component** or **Visualforce page** can be created for the custom user interface that displays the updates. It must **subscribe** to the **PushTopic channel**.

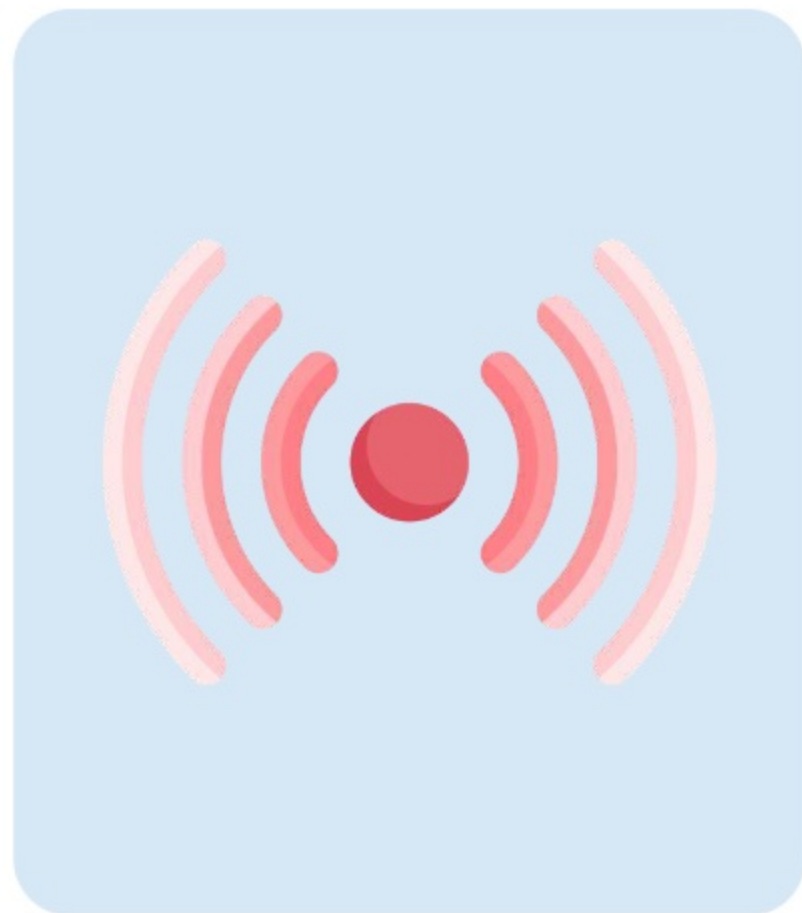


Streaming API

Salesforce Streaming API can be used to **stream notifications** to a Salesforce user interface and **update** it based on **data changes**. It comprises the components below.

❖ COMETD IMPLEMENTATION

A JavaScript-based implementation of the **Bayeux protocol (CometD)** is required to subscribe to the PushTopic channel. The **empApi** component in a **Lightning web component** or **Aura component** uses CometD and can be used to subscribe to a channel. A **Visualforce page** can utilize a **JavaScript library uploaded as a static resource** for subscription.



More on Streaming API

The Salesforce Streaming API consists of **two generations** of real-time notification solutions.

❖ 1ST-GENERATION

Salesforce developed and offered the following **initial** Streaming API products for meeting real-time notification requirements:

PushTopic events - push data to subscribers when records that qualify as a result of a SOQL query specified in a PushTopic record are created, updated, deleted, or undeleted.

Generic Streaming events - push notifications with data that is not tied to Salesforce record changes such as team announcements, events outside the Salesforce ecosystem, etc.

❖ 2ND-GENERATION

Then, Salesforce release the next generation of Streaming API solutions that provide more **flexibility, scalability, and security**:

Change Data Capture - the new version of PushTopic events that returns records of all supported changed fields and doesn't require a SOQL query specification.

Platform Events - the new version of Generic Streaming and supports publishing and subscribing to custom notifications, defining custom schema, monitoring activities, and more.

More on Streaming API

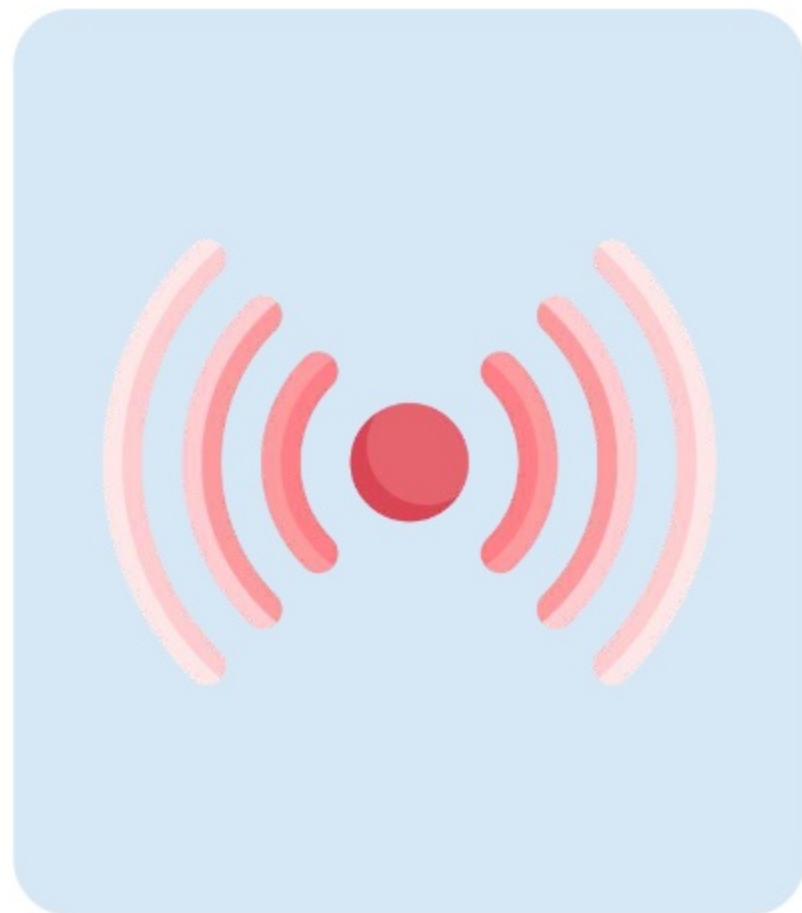
The next-generation Streaming API solutions, **platform events** and **change data capture**, provide the following capabilities.

❖ SUBSCRIPTION

Apex triggers can be used to subscribe to platform events and change data capture events. Also, **flows** and **processes** are capable of subscribing to platform events. For external systems, **CometD** or the **Pub/Sub API** can be used to subscribe to platform events.

❖ ENCRYPTION

Platform events and Change Data Capture allow encryption of **data at rest**, or data that is not moving across a network.



More on Streaming API

The next-generation Streaming API solutions, **platform events** and **change data capture**, provide the following capabilities.

❖ VERSIONED

Platform events and Change Data Capture supports **versioned schema**. Schema changes are detected and tracked using a **schema ID** so that event messages can be parsed accordingly.

❖ SUPPORT

Unlike its predecessor, Change Data Capture support **more objects** and **features** such as a header field which can be used to contain metadata related to the event.



Publish/Subscribe Solution

Table of Contents 

Platform Events

Platform Events can be utilized for **real-time delivery of notifications** in **Salesforce** or an **external system** triggered by **events in either system**. It comprises the components below.

❖ PLATFORM EVENT DEFINITION

A **platform event** can be defined in Salesforce Setup for a particular event that occurs in **Salesforce or an external system** by giving it a **name** and adding **custom fields**. For example, a platform event named **Opportunity_Event__e** can be created for opportunities that reach the 'Closed Won' stage. These custom field types are supported: **Checkbox**, **Date**, **Date/Time**, **Number**, **Text**, and **Text Area (Long)**

❖ EVENT PUBLISHER

Salesforce can **publish platform event messages** received by an external system using **Apex**, **Process Builder** or **Flow Builder**. An **external system** can **publish platform event messages** received by Salesforce using **Salesforce APIs**, such as **REST API**, **SOAP API**, or **Bulk API**.



Platform Events

Platform Events can be utilized for **real-time delivery of notifications** in **Salesforce** or an **external system** triggered by **events in either system**. It comprises the components below.

◆ EVENT SUBSCRIBER

Salesforce can subscribe to receive platform event messages using an **Apex Trigger, Lightning Web Component, Aura Component, Process Builder, or Flow Builder**. An **external system can subscribe** to receive platform event messages through **CometD (using a custom CometD client or EMP Connector)** or **Pub/Sub API**.



Pub/Sub API

[Table of Contents](#) 

Pub/Sub API

The **Pub/Sub API** is a secure and highly performant **gRPC-based API** that enables external applications to publish and subscribe to **platform events** in Salesforce.



gRPC Framework

The gRPC is an open-source **Remote Procedure Call (RPC)** framework developed by Google that enables streamlined communication between one application and another that is built on a different technology and/or hosted on a different environment.



Capabilities

An alternative to **CometD**, the **Pub/Sub API** enables an external system to **publish** and **subscribe** to platform events and **retrieve event schema** from Salesforce. It allows a subscriber also to define the **number of event messages** it will accept for regulated processing at a given time.

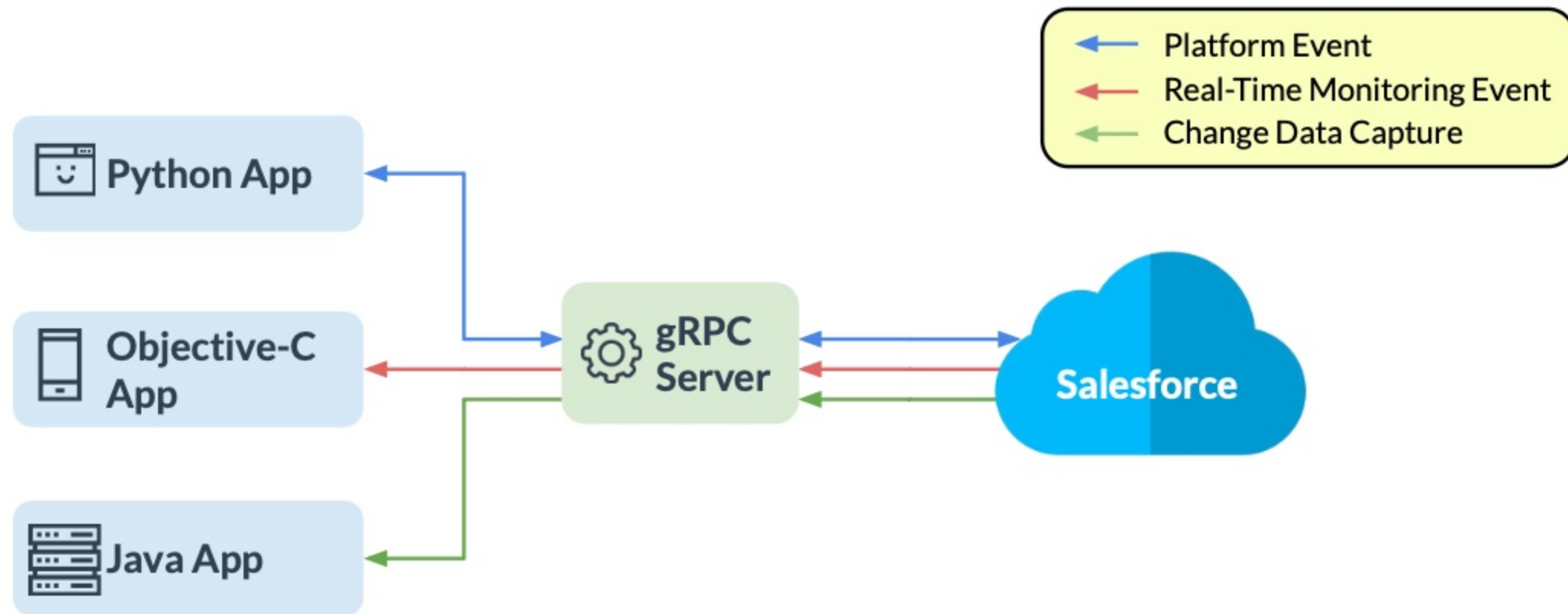


Supported Languages

Applications that are developed using **Python, Go, Java, Node.js, PHP, Ruby**, and more can use Pub/Sub API to **communicate with Salesforce** by receiving and sending Streaming API events.

Pub/Sub API

The Pub/Sub API supports **platform events**, **change data capture events**, and **real-time event monitoring events**.

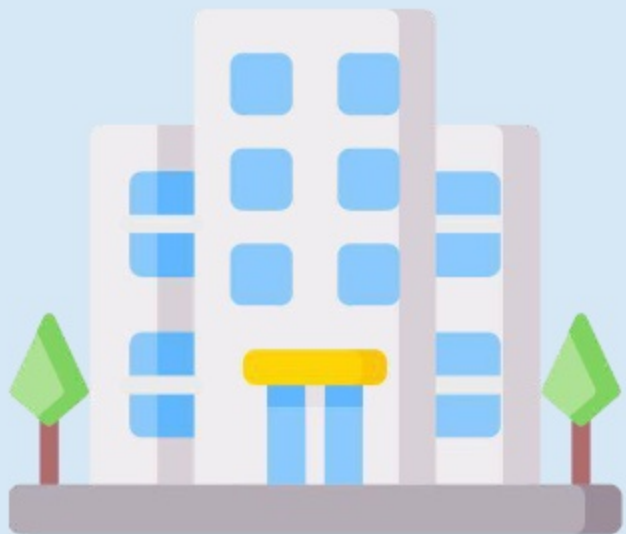


Business Scenario

Table of Contents 

Business Scenario: Part 1

Read the business scenario below and determine the most appropriate solutions for the company's requirements.

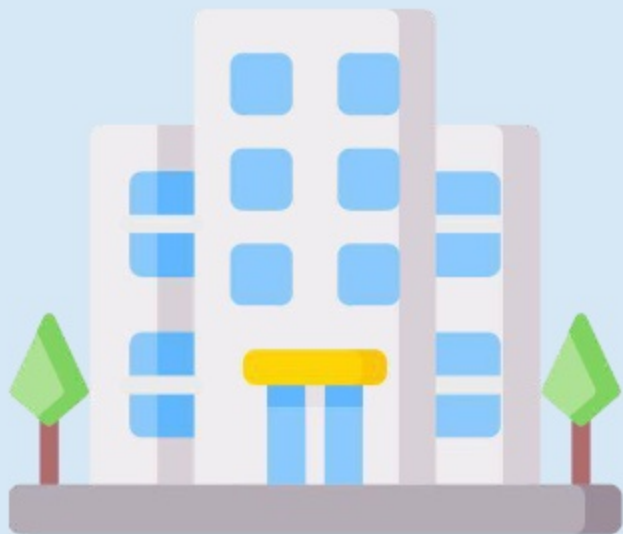


Cosmic Grocery is a company in the United States that sells grocery items through a public website and an app that can be downloaded by customers. Thousands of customers place orders every day. More than 10 million customers have purchased the company's products till date.

Salesforce is used by the company to store accounts, contacts, and products. Although it is also used for opportunity management, most of the opportunities are associated with businesses that want to purchase items in bulk. An enterprise resource planning (ERP) system contains tools that are used to gather and interpret data on employee training, recruiting, payroll, benefits, and talent management. A supply chain management (SCM) system allows the company to perform functions such as acquisition of products, supplier scheduling, inventory management, and inspection of products.

Business Scenario: Part 2

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



The executive management of the company recently reached a conclusion in a board meeting that the company would benefit greatly by making the business processes more efficient and reducing manual work by the company's employees who use the three main enterprise systems. It was determined that this objective could be achieved by integrating the systems.

As a result, a team of experts was created to gather and document the integration requirements by collaborating with various types of end users. The company has also hired a solution architect who needs to recommend suitable solutions to these requirements, including the components that are required to implement them.

Requirements & Solutions

[Table of Contents](#)

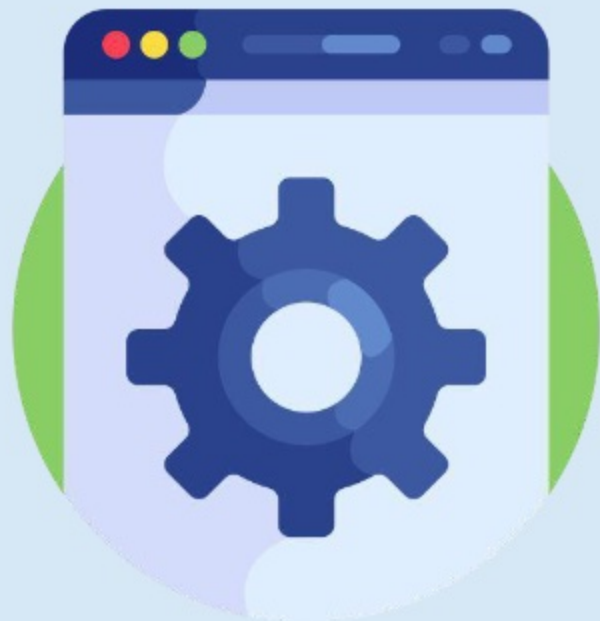


Requirement 1



SCENARIO

While managing opportunities and products in Salesforce, the sales reps of the company often need to check product inventory in the Supply Chain Management (SCM) system. To make the task less time consuming, the sales director has suggested the implementation of a feature that would allow the sales reps to access product inventory information from the record page of a product in Salesforce. A system administrator has informed the solution architect that the SCM system supports inbound REST API calls, which would be necessary to consider while designing a suitable solution.

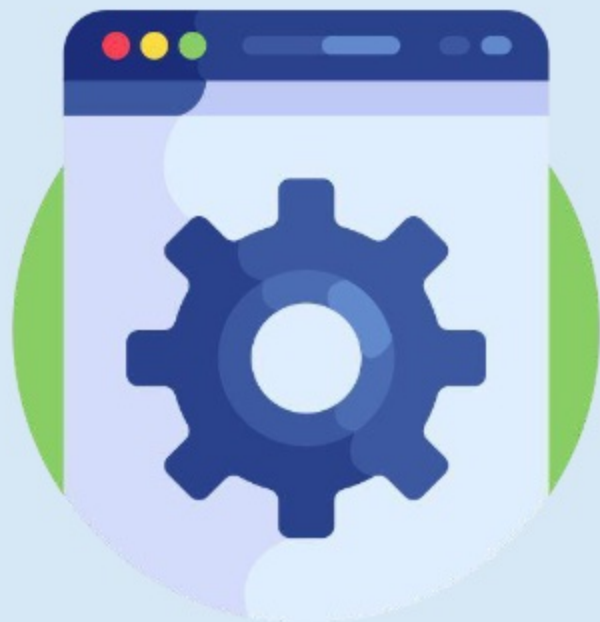


Solution 1

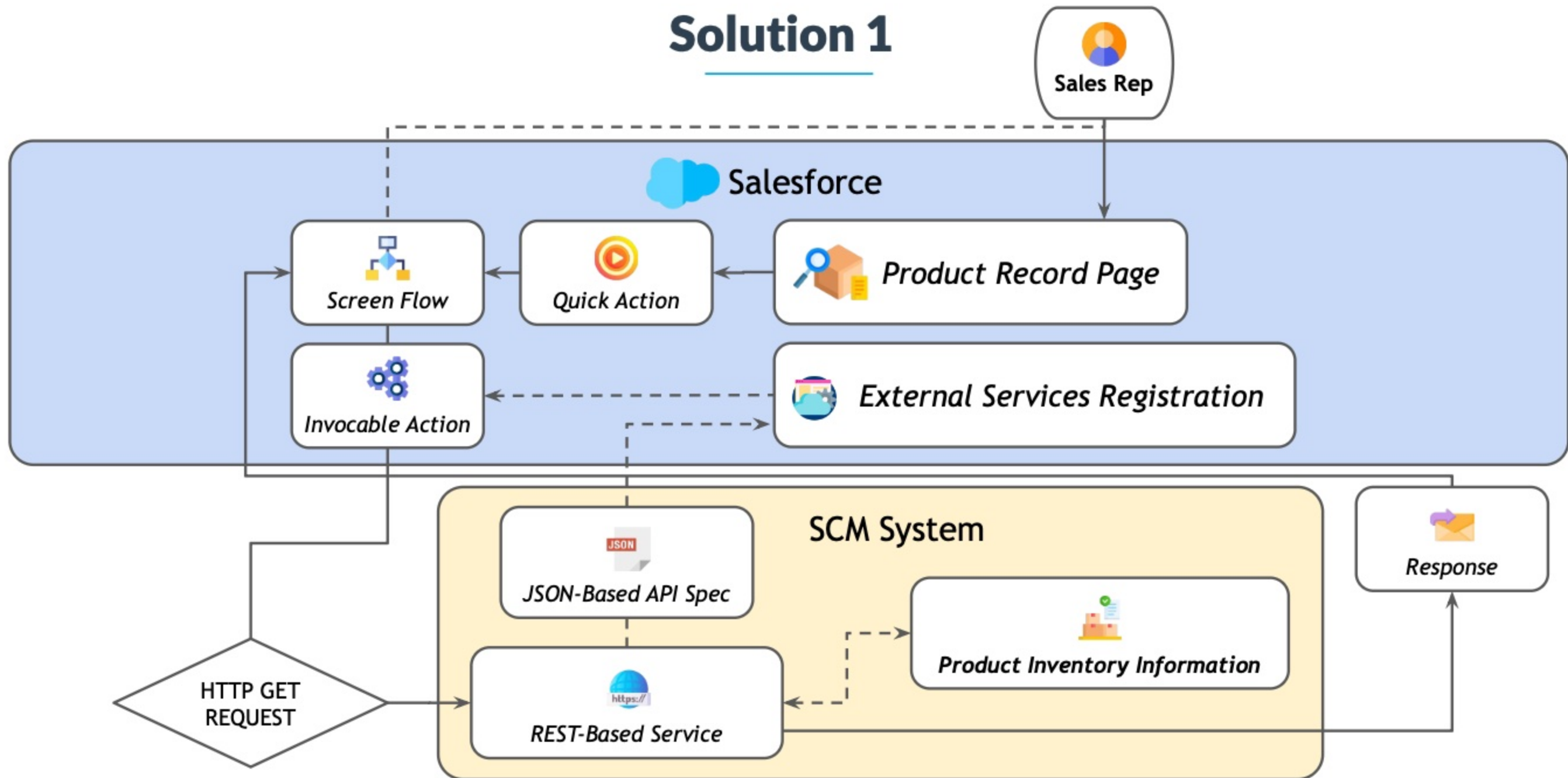


SOLUTION

The requirement can be met by using **External Services** in Salesforce. It would allow invoking a **REST-based service** hosted in the SCM system that sends inventory information about a particular product. The **API spec** of the service can be used for the registration of its operations with External Services in Salesforce Setup. The resulting **invocable action(s)** can be accessed in a **flow** built using **Flow Builder**. The flow can be associated with a **quick action** on the **Product record page**. Sales reps can use the quick action to invoke the flow and retrieve the product inventory information from the SCM system in real time.



Solution 1



Requirement 2



SCENARIO

The company has several warehouses that are used to store products. A custom object has been created in Salesforce to store warehouse data. But the Supply Chain Management (SCM) system is the primary system used for managing warehouses. When information about a warehouse is updated in the SCM, the corresponding warehouse record in Salesforce should be updated. Users in either system do not need to wait for the completion of the update.



Solution 2

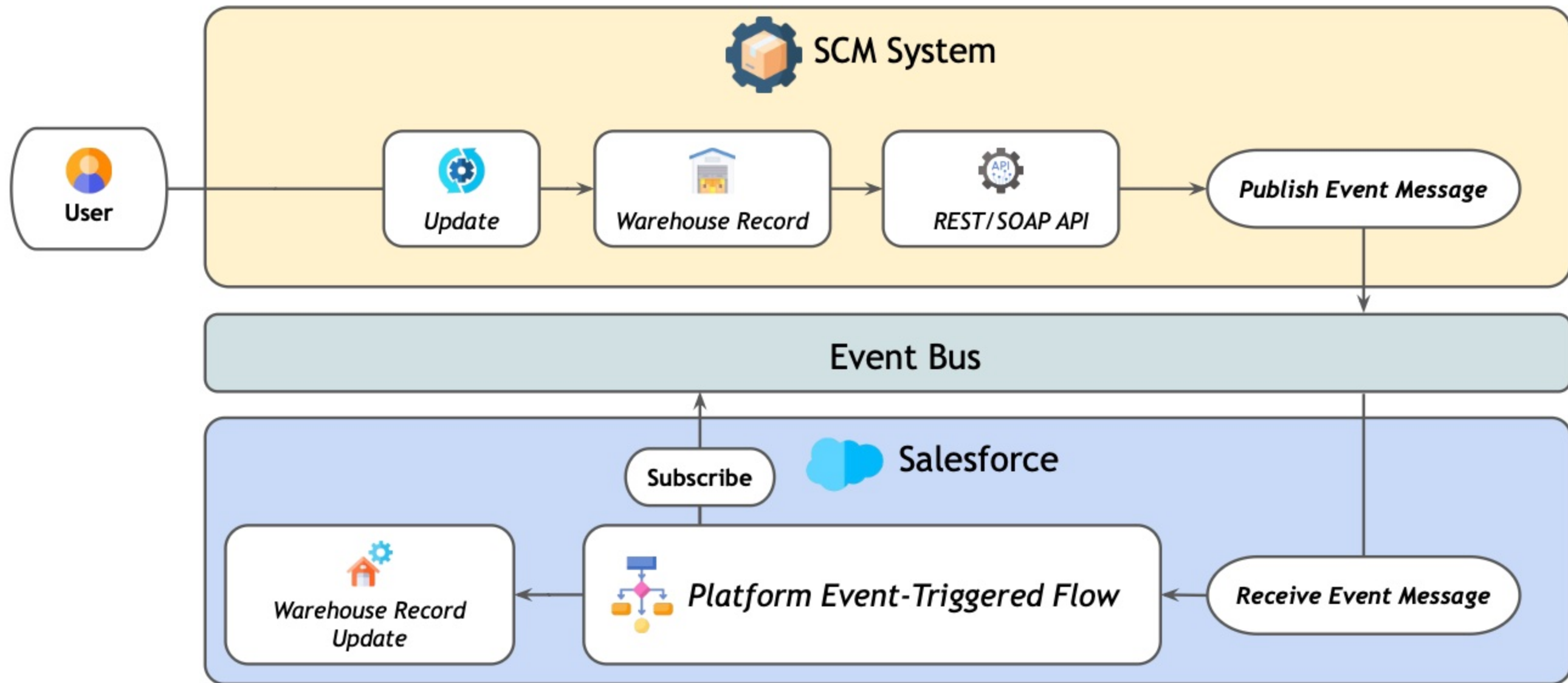


SOLUTION

A **platform event** can be defined for a warehouse update in Salesforce. **Custom fields** can be added to it for the warehouse data. When information about a warehouse is updated, the SCM system can publish a platform event message containing warehouse data using one of the **Salesforce APIs**, such as **REST API** or **SOAP API**. Salesforce can subscribe to incoming platform event notifications using a **platform event-triggered flow**. The flow can automatically update the corresponding warehouse record in Salesforce using the data in the platform event message.



Solution 2



Requirement 3



SCENARIO

Although the company uses Salesforce for account management, the Enterprise Resource Planning (ERP) system also contains account data. Sometimes sales users need to view customer accounts in the ERP system. However, when an account record in Salesforce is created, updated, deleted, or undeleted, a sales rep needs to manually perform the same operation in the ERP system. The sales director would like to automate the account data update in the ERP system based on the changes in Salesforce.



Solution 3

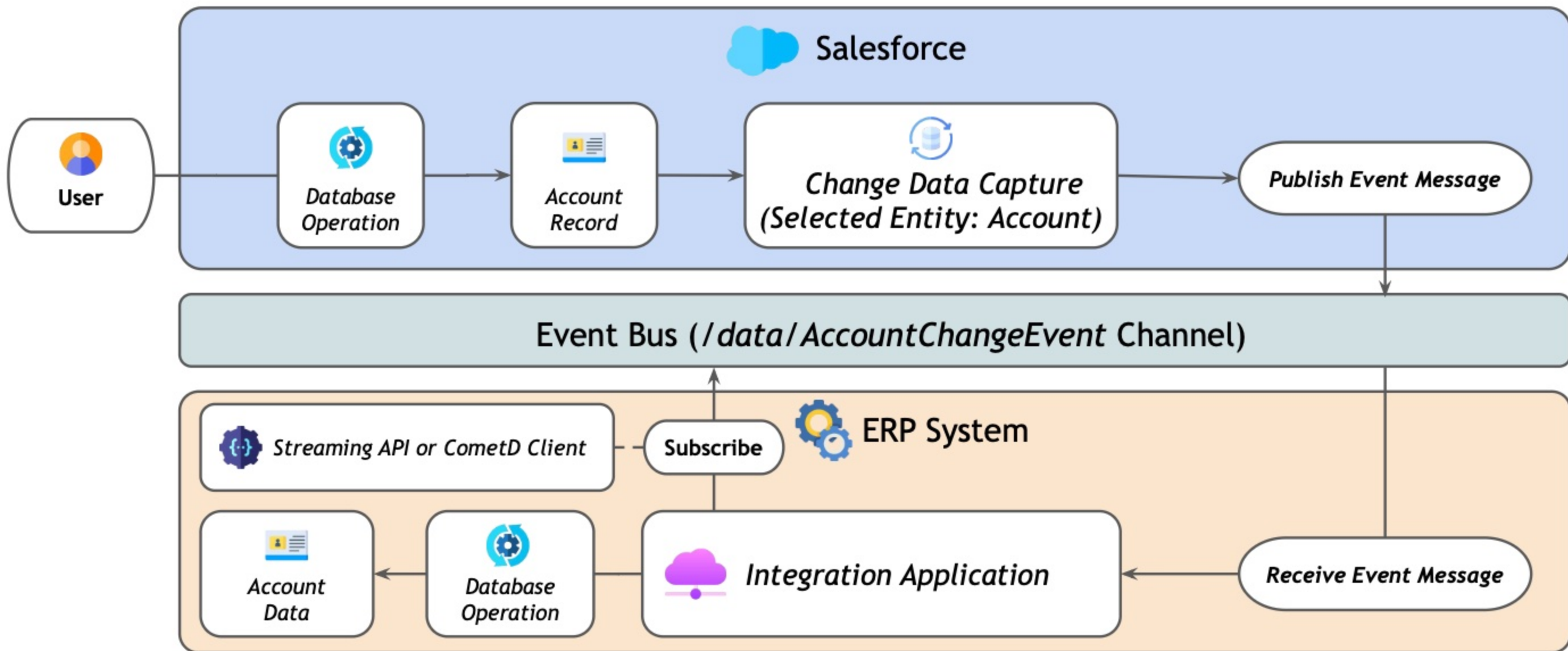


SOLUTION

Salesforce Change Data Capture can be used to meet this requirement. The **Account** object can be added to the list of **Selected Entities** on the **Change Data Capture** page in Setup to generate change event notifications when account records are **created, updated, deleted, or undeleted**. Using **Streaming API** or a **custom CometD client**, the ERP system can subscribe to the **/data/AccountChangeEvent** channel to receive change event notifications. Based on the data in a notification, an **integration app** in the ERP system can perform the necessary database operation to synchronize its account data with the record in Salesforce.



Solution 3

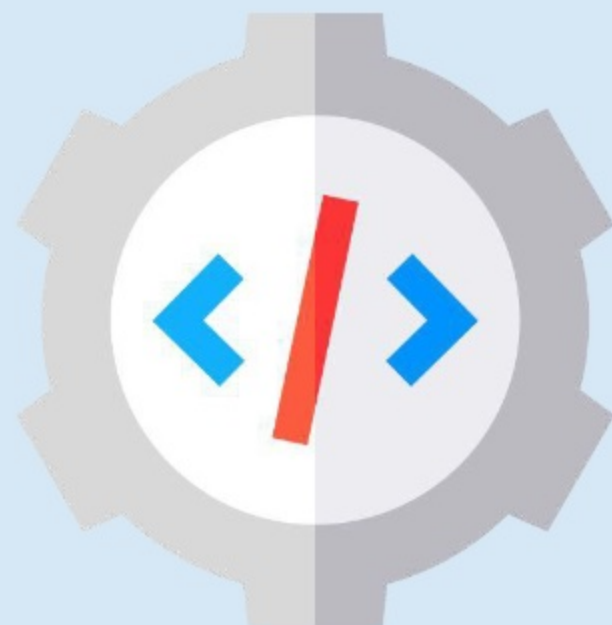


Requirement 4

SCENARIO



The company provides certain benefits to its employees, including health insurance, life insurance, retirement plan, tuition reimbursement, etc. An employee can even submit a request for a particular benefit. If an employee can access Salesforce, information about their benefits is stored on their user record to let the HRIS department easily access them in Salesforce. The department uses an app in the ERP system for benefits management and administration. When a new benefit needs to be added, an HRIS user should be able to quickly update the user record in Salesforce from the app's user interface. The User object in Salesforce has an External ID field that stores the unique identifier of each user in the ERP system.



Solution 4

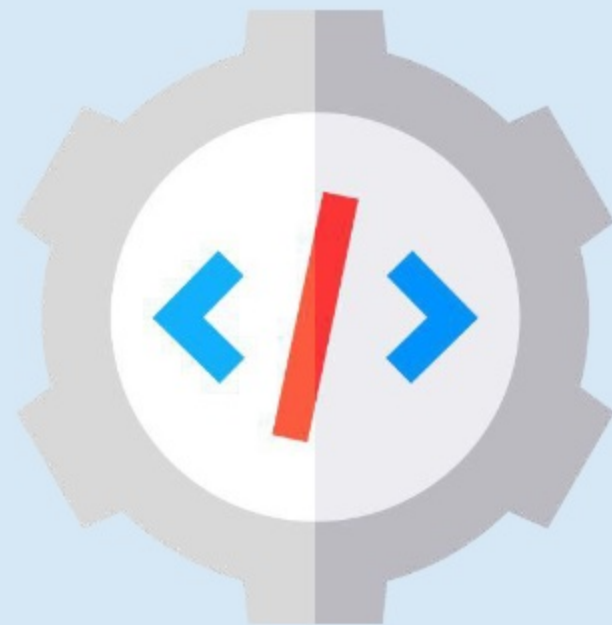


SOLUTION

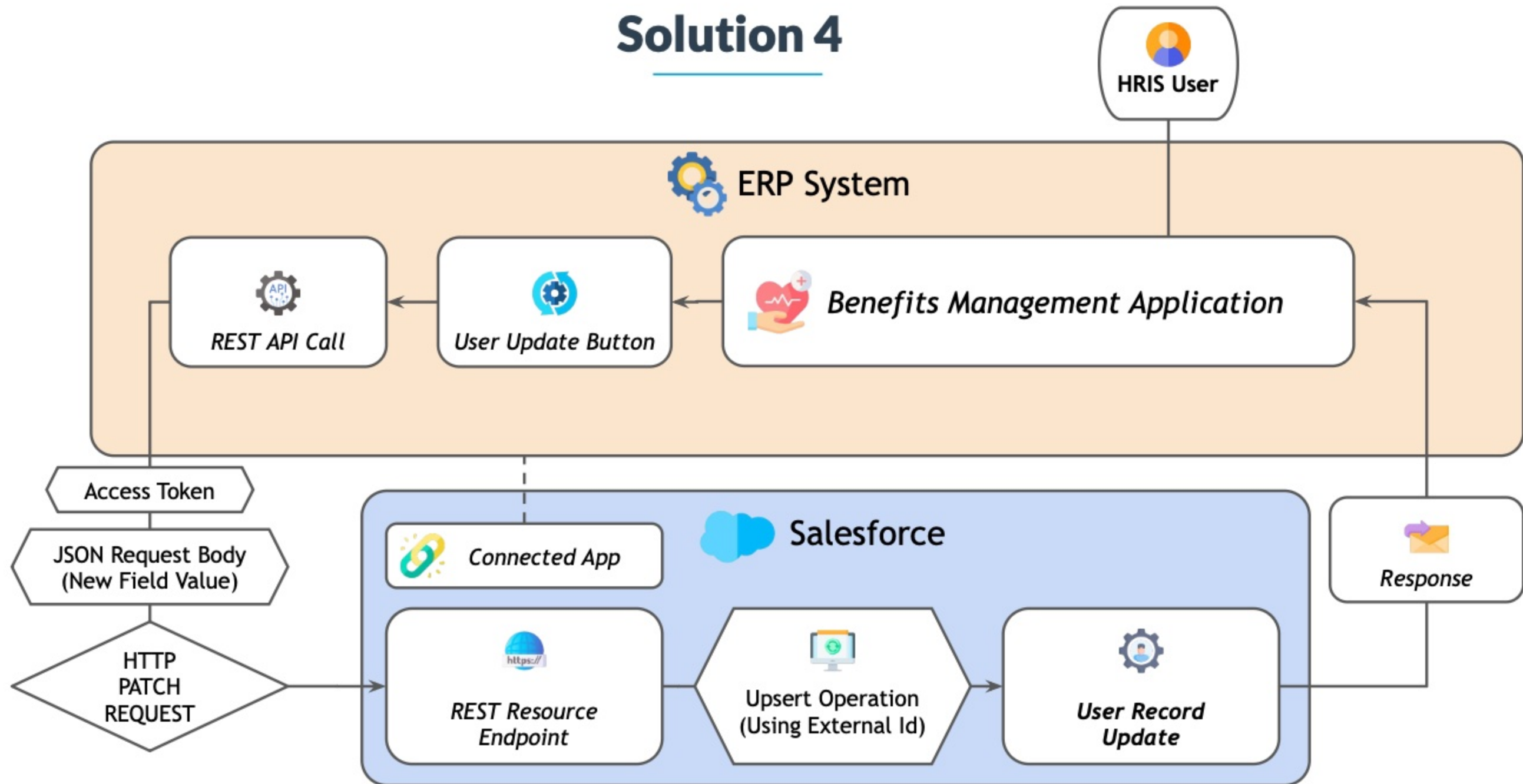
This requirement can be met by adding a **button** to the app's user interface that allows the current HRIS user to add a new benefit to a user record in Salesforce. The button can be associated with code that uses **REST API** call to **synchronously** update a user record in Salesforce. The **PATCH** method and the following endpoint can be used to perform an '**upsert**' operation using the **External Id** field on the User object:

services/data/v60.0/subjects/User/ExternalId__c/fieldValue

The **JSON request body** should contain the **new value of the field** that is used to store benefits. Also, a **connected app** should be created in Salesforce to authorize the ERP system and obtain the **access token** for REST API calls.



Solution 4



Learn More

-  [Integration Patterns and Practices](#)
-  [Application Integration Patterns for Salesforce Lightning Platform](#)
-  [Change Data Capture Basics](#)
-  [Platform Event Basics](#)
-  [Get Started with External Services](#)
-  [Get to Know the Salesforce Lightning Platform APIs](#)
-  [Upsert Records using External IDs](#)